Summary

Enuresis

Daytime enuresis

- Consider causes – developmental or psychogenic, bladder instability or neuropathy, urinary tract infection, constipation, ectopic ureter.

Secondary (onset) enuresis

- Consider – emotional upset, UTI, polyuria from an osmotic diuresis in diabetes mellitus or a renal concentrating disorder.

Proteinuria

Transient proteinuria may occur during febrile illnesses or after exercise and does not require investigation. Persistent proteinuria is significant and should be quantified by measuring the urine protein/creatinine ratio in an early morning sample (protein should not exceed 20 mg/mmol of creatinine).

A common cause is orthostatic (postural) proteinuria, where proteinuria is only found when the child is upright, i.e. during the day. It can be diagnosed by measuring the urine protein/creatinine ratio in a series of early morning urine specimens. The prognosis is excellent and further investigations are not necessary.

Box 18.2 Causes of proteinuria

- Orthostatic proteinuria
- Glomerular abnormalities
  - Minimal change disease
  - Glomerulonephritis
  - Abnormal glomerular basement membrane (familial nephritides)
- Increased glomerular filtration pressure
- Reduced renal mass
- Hypertension
- Tubular proteinuria.

Other causes of proteinuria, which needs further evaluation, are listed in Box 18.2.

Nephrotic syndrome

In nephrotic syndrome, heavy proteinuria results in a low plasma albumin and oedema. The cause of the condition is unknown, but a few cases are secondary to systemic diseases such as Henoch–Schönlein purpura (HSP) and other vasculitides, e.g. systemic lupus erythematosus (SLE), infections (e.g. malaria) or allergens (e.g. bee sting).

Clinical signs of the nephrotic syndrome are:

- Periorbital oedema (particularly on waking), the earliest sign (Fig. 18.16)
- Scrotal or vulval, leg and ankle oedema (Fig. 18.17)
- Ascites
- Breathlessness due to pleural effusions and abdominal distension.

The initial investigations are listed in Box 18.3.

Steroid-sensitive nephrotic syndrome

In 85–90% of children with nephrotic syndrome, the proteinuria resolves with corticosteroid therapy (steroid-sensitive nephrotic syndrome). These children do not progress to renal failure. It is commoner in boys than in girls, in Asian children than in Caucasians, and there is a weak association with atopy. It is often

Box 18.3 Investigations performed at presentation of nephrotic syndrome

- Urine protein – on test strips ('dipstick')
- Full blood count and ESR
- Urea, electrolytes, creatinine, albumin
- Complement levels – C3, C4
- Antistreptolysin O or anti-DNAse B titres and throat swab
- Urine microscopy and culture
- Urinary sodium concentration
- Hepatitis B and C screen
- Malaria screen if travel abroad.

Nephrotic syndrome



Figure 18.16 Facial oedema in nephrotic syndrome.



Figure 18.17 Gross oedema of the scrotum and legs as well as abdominal distension from ascites.

precipitated by respiratory infections. Features suggesting steroid-sensitive nephrotic syndrome are:

- Age between 1 and 10 years
- No macroscopic haematuria
- Normal blood pressure
- Normal complement levels
- Normal renal function.

Management

The most widely used protocol is to initially give oral corticosteroids (60 mg/m^2 per day of prednisolone), unless there are atypical features. After 4 weeks, the dose is reduced to 40 mg/m^2 on alternate days for 4 weeks and then stopped. The median time for the urine to become free of protein is 11 days. However, there is now good evidence that extending the initial course of steroids, by gradually tapering the alternate day part of the course, leads to a marked reduction in the proportion of children who develop a frequently relapsing or steroid-dependent course, and this scheme is increasingly being adopted. Children who do not respond to 4–8 weeks of corticosteroid therapy or have atypical features may have a more complex diagnosis and require a renal biopsy. Renal histology in steroid-sensitive nephrotic syndrome is usually normal on light microscopy but fusion of the specialised epithelial cells that invest the glomerular capillaries (podocytes) is seen on electron microscopy. For this reason, it is called minimal change disease.

The child with nephrotic syndrome is susceptible to several serious complications at presentation or relapse:

- *Hypovolaemia*. During the initial phase of oedema formation the intravascular compartment may become volume depleted. The child who becomes hypovolaemic characteristically complains of abdominal pain and may feel faint. There is peripheral vasoconstriction and urinary sodium retention. A low urinary sodium (<20 mmol/L) and a high packed cell volume of red blood cells are indications of hypovolaemia, which requires urgent treatment with intravenous albumin as the child is at risk of vascular thrombosis and shock. Increasing peripheral oedema, assessed clinically and by daily weight, may cause discomfort and respiratory compromise. If severe, this may need treatment with intravenous albumin. Care must be taken with the use of colloid, as it may precipitate pulmonary oedema and hypertension from fluid overload, and also with diuretics, which may cause or worsen hypovolaemia.
- *Thrombosis*. A hypercoagulable state, due to urinary losses of antithrombin, thrombocytosis which may be exacerbated by steroid therapy, increased synthesis of clotting factors and increased blood viscosity from the raised haematocrit, predisposes to thrombosis. This may affect the brain, limbs and splanchnic circulation with potentially catastrophic results.
- *Infection*. Children in relapse are at risk of infection with capsulated bacteria, especially *Pneumococcus*. Spontaneous peritonitis may occur. Pneumococcal and seasonal influenza vaccination is widely recommended. Chickenpox and shingles should be treated with aciclovir.
- *Hypercholesterolaemia*. This correlates inversely with the serum albumin, but the cause of the hyperlipidaemia is not fully understood.

Prognosis

This is summarised in Figure 18.18. Relapses are identified by parents on urine testing. The side-effects of corticosteroid therapy may be reduced by an alternate-day regimen. If relapses are frequent, or if a high maintenance dose is required, involvement of a paediatric nephrologist is advisable as other drug therapy may be considered to enable reduction in steroid use. Possible steroid-sparing agents include the immunomodulator levamisole, alkylating agents (e.g. cyclophosphamide), calcineurin inhibitors such as tacrolimus and ciclosporin A and the immunosuppressant mycophenolate mofetil.

Steroid-resistant nephrotic syndrome

(Table 18.4)

These children should be referred to a paediatric nephrologist. Management of the oedema is by diuretic therapy, salt restriction, ACE inhibitors and sometimes

Summary

Nephrotic syndrome

- Clinical signs – oedema (periorbital, scrotal or vulval, leg and ankle oedema, ascites, pleural effusions)
- Diagnosis – heavy proteinuria and low plasma albumin.

Steroid-sensitive nephrotic syndrome

- Characteristic features – 1 to 10 years old, no macroscopic haematuria, and blood pressure, complement levels and renal function are normal
- Management – oral corticosteroids, renal biopsy if unresponsive or atypical features
- Complications – hypovolaemia, thrombosis, infection (pneumococcal), hypercholesterolaemia
- Prognosis – may resolve or else there may be infrequent or frequent relapse.

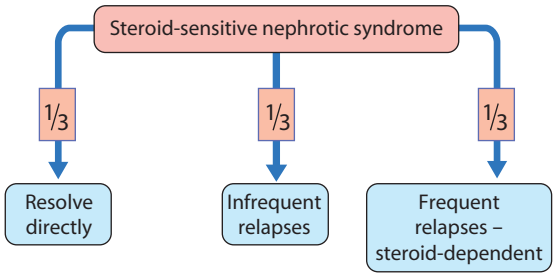


Figure 18.18 Clinical course in steroid-responsive nephrotic syndrome.

Table 18.4 Steroid-resistant nephrotic syndrome

Cause	Specific features	Prognosis
Focal segmental glomerulosclerosis	Most common Familial or idiopathic	30% progress to end-stage renal failure in 5 years; 20% respond to cyclophosphamide, ciclosporin, tacrolimus or rituximab Recurrence post-transplant is common
Mesangiocapillary glomerulonephritis (membranoproliferative glomerulonephritis)	More common in older children Haematuria and low complement level present	Decline in renal function over many years
Membranous nephropathy	Associated with hepatitis B May precede SLE	Most remit spontaneously within 5 years

NSAIDs (non-steroidal anti-inflammatory drugs), which may reduce proteinuria.

Congenital nephrotic syndrome

Congenital nephrotic syndrome presents in the first 3 months of life. It is rare. The commonest kind is recessively inherited and the gene frequency is particularly high in Finns. In the UK, it is more common in consanguineous families. It is associated with a high mortality, usually due to complications of hypoalbuminaemia rather than renal failure. The albuminuria is so severe that unilateral nephrectomy may be necessary for its control, followed by dialysis for renal failure, which is continued until the child is large and fit enough for renal transplantation.

An oedematous child – test for proteinuria to diagnose nephrotic syndrome.

Haematuria

Urine that is red in colour or tests positive for haemoglobin on urine sticks should be examined under the microscope to confirm haematuria (>10 red blood cells per high-power field). Glomerular haematuria is suggested by brown urine, the presence of deformed red cells (which occurs as they pass through the basement membrane) and casts, and is often accompanied by proteinuria. Lower urinary tract haematuria is usually red, occurs at the beginning or end of the urinary stream, is not accompanied by proteinuria and is unusual in children.

Urinary tract infection is the most common cause of haematuria (Box 18.4), although seldom as the only symptom. The history and examination may suggest the diagnosis, e.g. a family history of stone formation or nephritis or a history of trauma. A plan of investigation is outlined in Box 18.5.

A renal biopsy may be indicated if:

- There is significant persistent proteinuria.
- There is recurrent macroscopic haematuria
- Renal function is abnormal
- The complement levels are persistently abnormal.

Acute nephritis

The causes of acute nephritis in childhood are listed in Box 18.6. Increased glomerular cellularity restricts glomerular blood flow and therefore filtration is decreased. This leads to:

- decreased urine output and volume overload
- hypertension, which may cause seizures
- oedema, characteristically around the eyes
- haematuria and proteinuria.

Management is by attention to both water and electrolyte balance and the use of diuretics when necessary. Rarely, there may be a rapid deterioration in renal function (rapidly progressive glomerulonephritis). This may occur with any cause of acute nephritis, but is uncommon when the cause is post-streptococcal. If left untreated, irreversible renal failure may occur over weeks or months, so renal biopsy and subsequent treatment with immunosuppression and plasma exchange may be necessary.

Post-streptococcal and post-infectious nephritis

Usually follows a streptococcal sore throat or skin infection and is diagnosed by evidence of a recent streptococcal infection (culture of the organism, raised ASO/anti-DNAse B titres) and low complement C3 levels that return to normal after 3–4 weeks. Streptococcal nephritis is a common condition in developing countries, but has become uncommon in developed countries. Long-term prognosis is good.

Henoch–Schönlein purpura

Henoch–Schönlein purpura is the combination of some of the following features:

- Characteristic skin rash
- Arthralgia
- Periarticular oedema
- Abdominal pain
- Glomerulonephritis.

It usually occurs between the ages of 3 and 10 years, is twice as common in boys, peaks during the winter months and is often preceded by an upper respiratory

Haematuria

Box 18.4 Causes of haematuria

Non-glomerular

- Infection (bacterial, viral, TB, schistosomiasis)
- Trauma to genitalia, urinary tract or kidneys
- Stones
- Tumours
- Sickle cell disease
- Bleeding disorders
- Renal vein thrombosis
- Hypercalciuria.

Glomerular

- Acute glomerulonephritis (usually with proteinuria)
- Chronic glomerulonephritis (usually with proteinuria)
- IgA nephropathy
- Familial nephritis, e.g. Alport syndrome
- Thin basement membrane disease.

Box 18.5 Investigation of haematuria

All patients

- Urine microscopy (with phase contrast) and culture
- Protein and calcium excretion
- Kidney and urinary tract ultrasound
- Plasma urea, electrolytes, creatinine, calcium, phosphate, albumin
- Full blood count, platelets, clotting screen, sickle cell screen.

If suggestive of glomerular haematuria

- ESR, complement levels and anti-DNA antibodies
- Throat swab and antistreptolysin O/anti-DNAse B titres
- Hepatitis B and C screen
- Renal biopsy if indicated
- Test mother's urine for blood (if Alport syndrome suspected)
- Hearing test (if Alport syndrome suspected).

Box 18.6 Causes of acute nephritis

- Post-infectious (including streptococcus)
- Vasculitis (Henoch–Schönlein purpura or, rarely, SLE, Wegener granulomatosis, microscopic polyarteritis, polyarteritis nodosa)
- IgA nephropathy and mesangiocapillary glomerulonephritis
- Anti-glomerular basement membrane disease (Goodpasture syndrome) – very rare.

infection. Despite much research, the cause is unknown. It is postulated that genetic predisposition and antigen exposure increase circulating IgA levels and disrupt IgG synthesis. The IgA and IgG interact to produce complexes that activate complement and are deposited in affected organs, precipitating an inflammatory response with vasculitis.

Clinical findings (Fig. 18.19)

At presentation, affected children often have a fever. The *rash* is the most obvious feature. It is symmetrically distributed over the buttocks, the extensor surfaces of the arms and legs, and the ankles. The trunk is spared unless lesions are induced by trauma. The rash may initially be urticarial, rapidly becoming maculopapular and purpuric, is characteristically palpable and may recur over several weeks. The rash is the first clinical feature in about 50% and is the cornerstone of the diagnosis, which is clinical.

Joint pain occurs in two-thirds of patients, particularly of the knees and ankles. There is *periarticular oedema*. Long-term damage to the joints does not occur, and symptoms usually resolve before the rash goes.

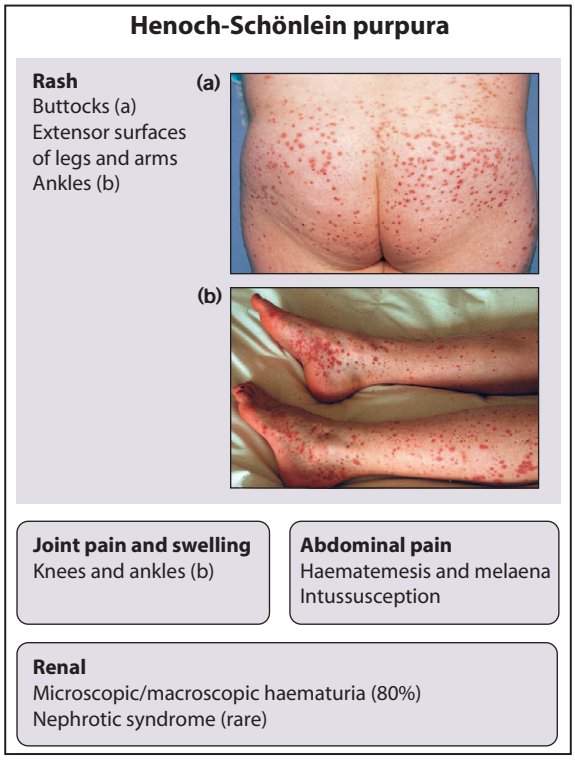


Figure 18.19 Main clinical manifestations of Henoch–Schönlein purpura. **(a)** Rash on buttocks. (Courtesy of Dr Michael Markiewicz.) **(b)** Rash around the extensor surfaces of the legs and slight joint swelling. (Courtesy of Professor Tauny Southwood.)

Colicky abdominal pain occurs in many children and, if severe, can be treated with corticosteroids. Gastrointestinal petechiae can cause haematemesis and melaena. Intussusception can occur and can be particularly difficult to diagnose under these circumstances. Ileus, protein-losing enteropathy, orchitis and occasionally central nervous system involvement are rare complications.

Renal involvement is common, but is rarely the first symptom. Over 80% have microscopic or macroscopic haematuria or mild proteinuria. These children usually make a complete recovery. If proteinuria is more severe, nephrotic syndrome may result. Risk factors for progressive renal disease are heavy proteinuria, oedema, hypertension and deteriorating renal function, when a renal biopsy will determine if treatment is necessary. All children with renal involvement are followed for a year to detect those with persisting urinary abnormalities (5–10%), who require long-term follow-up. This is necessary as hypertension and declining renal function may develop after an interval of several years.

IgA nephropathy

This may present with episodes of macroscopic haematuria, commonly in association with upper respiratory tract infections. Histological findings and management are as for Henoch–Schönlein purpura, which may be a variant of the same pathological process but not restricted to the kidney. The prognosis in children is better than that in adults.

Familial nephritis

The commonest familial nephritis is Alport syndrome. This is usually an X-linked recessive disorder that progresses to end-stage renal failure by early adult life in males and is associated with nerve deafness and ocular defects. The mother may have haematuria.

Vasculitis

The commonest vasculitis to involve the kidney is Henoch–Schönlein purpura (see above). However, renal involvement may occur in rarer vasculitides such as polyarteritis nodosa, microscopic polyarteritis and Wegener granulomatosis. Characteristic symptoms are fever, malaise, weight loss, skin rash and arthropathy with prominent involvement of the respiratory tract in Wegener disease. ANCA (antineutrophil cytoplasm antibodies) are present and diagnostic in these diseases. Renal arteriography, to demonstrate the presence of aneurysms, will diagnose polyarteritis nodosa. Renal involvement may be severe and rapidly progressive. Treatment is with steroids, plasma exchange and intravenous cyclophosphamide, which may need to be continued for many months.

Systemic lupus erythematosus (SLE)

SLE is a disease that presents mainly in adolescent girls and young women. It is much commoner in Asians and Afro-Caribbeans than Caucasians. It is characterised by the presence of multiple autoantibodies, including antibodies to double-stranded DNA. The C3 and C4 components of complement may be low, particularly during active phases of the disease. Haematuria and proteinuria are indications for renal biopsy, as immunosuppression is always necessary and its intensity will depend on the severity of renal involvement.

Summary

Acute nephritis

- Cause: usually post-infectious or follows a streptococcal infection, but also vasculitis (including Henoch–Schönlein purpura), IgA nephropathy and familial nephritis
- Clinical features: oedema (around the eyes), hypertension, decreased urine output, haematuria and proteinuria
- Management: fluid and electrolyte balance, diuretics, monitor for rapid deterioration in renal function.

Hypertension

Blood pressure in children needs to be measured with a cuff over two-thirds the length of the upper arm (see Ch. 2). Blood pressure increases with age and height and readings should be plotted on a centile chart (see Appendix). Hypertension is blood pressure above 95th percentile for height, age and sex. Symptomatic hypertension in children is usually secondary to renal, cardiac or endocrine causes (Box 18.7).

Presentation includes vomiting, headaches, facial palsy, hypertensive retinopathy, convulsions or proteinuria. Failure to thrive and cardiac failure are the most common features in infants. Phaeochromocytoma may also cause paroxysmal palpitations and sweating.

Some causes are correctable, e.g. nephrectomy for unilateral scarring, angioplasty for renal artery stenosis, surgical repair of coarctation of the aorta, resection of a phaeochromocytoma, but in most cases medical treatment is necessary with antihypertensive drugs.

Box 18.7 Causes of hypertension

- **Renal**
  - Renal parenchymal disease
  - Renovascular, e.g. renal artery stenosis
  - Polycystic kidney disease (ARPKD and ADPKD)
  - Renal tumours
- **Coarctation of the aorta**
- **Catecholamine excess**
  - Phaeochromocytoma
  - Neuroblastoma
- **Endocrine**
  - Congenital adrenal hyperplasia
  - Cushing syndrome or corticosteroid therapy
  - Hyperthyroidism
- **Essential hypertension**
  - A diagnosis of exclusion.

Box 18.8 Causes of palpable kidneys

Unilateral

- Multicystic kidney
- Compensatory hypertrophy
- Obstructed hydronephrosis
- Renal tumour (Wilms tumour)
- Renal vein thrombosis

Bilateral

- Autosomal recessive (infantile) polycystic kidneys
- Autosomal dominant (adult) polycystic kidneys
- Tuberous sclerosis
- Renal vein thrombosis.

Early detection of hypertension is important. Any children with a renal abnormality should have their blood pressure checked annually throughout life. Children with a family history of essential hypertension should be encouraged to restrict their salt intake, avoid obesity and have their blood pressure checked regularly.

Renal masses

An abdominal mass identified on palpating the abdomen should be investigated promptly by ultrasound scan. The causes of palpable kidneys are shown in Box 18.8. Bilaterally enlarged kidneys in early life are most frequently due to autosomal recessive polycystic kidney disease, which is associated with hypertension, hepatic fibrosis and progression to chronic renal failure. This form of polycystic kidney disease must be distinguished from the autosomal dominant adult-type polycystic kidney disease, which has a more benign prognosis in childhood with onset of renal failure in adulthood.

Renal calculi

Renal stones are uncommon in childhood (Fig. 18.20). When they occur, predisposing causes must be sought:

- Urinary tract infection
- Structural anomalies of the urinary tract
- Metabolic abnormalities.

The commonest are phosphate stones associated with infection, especially with *Proteus*. Calcium-containing stones occur in idiopathic hypercalciuria, the most common metabolic abnormality, and with increased urinary urate and oxalate excretion. Deposition of calcium in the parenchyma (nephrocalcinosis) may occur with hypercalciuria, hyperoxaluria and distal renal tubular acidosis. Nephrocalcinosis may be a complication of furosemide therapy in the neonate. Cystine and xanthine stones are rare.

Presentation may be with haematuria, loin or abdominal pain, UTI or passage of a stone.

Stones that are not passed spontaneously should be removed, by either lithotripsy or surgery, and any predisposing structural anomaly repaired. A high fluid intake is recommended in all affected children. If the cause is a metabolic abnormality, specific therapy may be possible.

Box 18.9 Causes of Fanconi syndrome

Idiopathic

Secondary to inborn errors of metabolism

- Cystinosis (an autosomal recessive disorder causing intracellular accumulation of cystine)
- Glycogen storage disorders
- Lowe syndrome (oculocerebrorenal dystrophy)
- Galactosaemia
- Fructose intolerance
- Tyrosinaemia
- Wilson disease

Acquired

- Heavy metals
- Drugs and toxins
- Vitamin D deficiency.

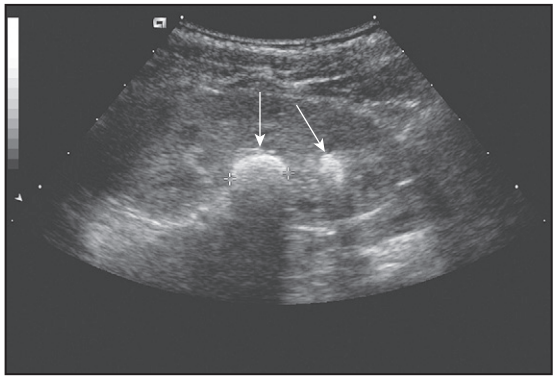


Figure 18.20 Renal ultrasound showing a staghorn calculus.

Renal tubular disorders

Abnormalities of renal tubular function may occur at any point along the length of the nephron and affect any of the substances handled by it.

Generalised proximal tubular dysfunction (Fanconi syndrome)

Proximal tubule cells are among the most metabolically active in the body, so are especially vulnerable to cellular damage. The cardinal features are excessive urinary loss of amino acids, glucose, phosphate, bicarbonate, sodium, calcium, potassium and urate. The causes are listed in Box 18.9. Fanconi syndrome should be considered in a child presenting with:

- Polydypsia and polyuria
- Salt depletion and dehydration
- Hyperchloraemic metabolic acidosis
- Rickets
- Failure to thrive/poor growth.

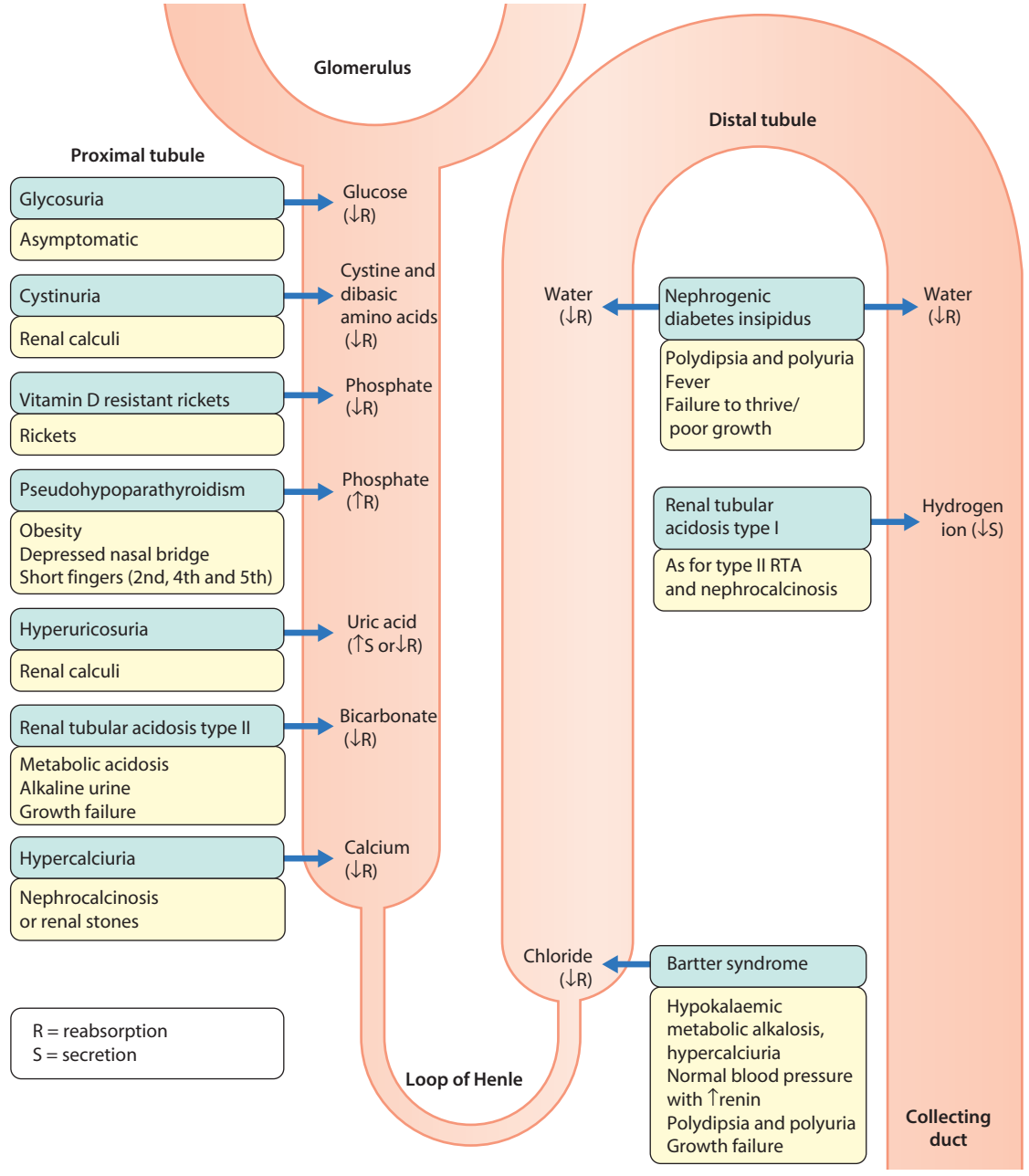


Figure 18.21 Schematic diagram of specific transport defects in some renal tubular disorders.

Specific transport defects

See Figure 18.21.

Acute kidney injury

Acute kidney injury has acute renal failure at the most severe end of the spectrum where there is a sudden, potentially reversible, reduction in renal function.

Oliguria (<0.5 ml/kg per hour) is usually present. It can be classified as (see Box 18.10):

- Prerenal: the commonest cause in children
- Renal: there is salt and water retention; blood, protein and casts are often present in the urine; and there may be symptoms specific to an accompanying disease (e.g. haemolytic uraemic syndrome)
- Postrenal: from urinary obstruction.

Box 18.10 Causes of acute renal failure

Prerenal

- Hypovolaemia:
  - Gastroenteritis
  - Burns
  - Sepsis
  - Haemorrhage
  - Nephrotic syndrome
- Circulatory failure

Renal

- Vascular:
  - Haemolytic uraemic syndrome (HUS)
  - Vasculitis
  - Embolus
  - Renal vein thrombosis
- Tubular:
  - Acute tubular necrosis (ATN)
  - Ischaemic
  - Toxic
  - Obstructive
- Glomerular:
  - Glomerulonephritis
- Interstitial:
  - Interstitial nephritis
  - Pyelonephritis

Postrenal

- Obstruction:
  - Congenital e.g. posterior urethral valves
  - Acquired e.g. blocked urinary catheter

Acute-on-chronic renal failure is suggested by the child having growth failure, anaemia and disordered bone mineralisation (renal osteodystrophy).

Management

Children with acute renal failure should have their circulation and fluid balance meticulously monitored. Investigation by ultrasound scan will identify obstruction of the urinary tract, the small kidneys of chronic renal failure, or large, bright kidneys with loss of cortical medullary differentiation typical of an acute process.

Prerenal failure

This is suggested by hypovolaemia. The fractional excretion of sodium is very low as the body tries to retain fluid. The hypovolaemia needs to be urgently corrected with fluid replacement and circulatory support if acute tubular necrosis is to be avoided.

Renal failure

If there is circulatory overload, restriction of fluid intake and challenge with a diuretic may increase urine output sufficiently to allow gradual correction of sodium and water balance. A high-calorie, normal protein feed will decrease catabolism, uraemia and hyperkalaemia. Emergency management of metabolic acidosis, hyperkalaemia and hyperphosphataemia is shown in Table 18.5. If the cause of renal failure is not obvious, a renal biopsy should be performed to identify rapidly progressive glomerulonephritis, as this may need immediate treatment with immunosuppression. The two commonest renal causes of acute renal failure in children in the UK are the haemolytic uraemic syndrome and acute tubular necrosis, the latter usually in the setting of multisystem failure in the intensive care unit or following cardiac surgery.

Postrenal failure

This requires assessment of the site of obstruction and relief by nephrostomy or bladder catheterisation. Surgery can be performed once fluid volume and electrolyte abnormalities have been corrected.

Table 18.5 Some metabolic abnormalities in acute renal failure and their therapy

Metabolic abnormality	Treatment
Metabolic acidosis	Sodium bicarbonate
Hyperphosphataemia	Calcium carbonate
	Dietary restriction
Hyperkalaemia	Calcium gluconate if ECG changes
	Salbutamol (nebulised or intravenous)
	Calcium exchange resin
	Glucose and insulin
	Dietary restriction
	Dialysis

Dialysis

Dialysis in acute renal failure is indicated when there is:

- Failure of conservative management
- Hyperkalaemia
- Severe hypo- or hypernatraemia
- Pulmonary oedema or hypertension
- Severe acidosis
- Multisystem failure.

Peritoneal dialysis or haemodialysis can be undertaken for acute renal failure. If plasma exchange is part of the treatment, haemodialysis is used. If there is cardiac decompensation or hypercatabolism, continuous arteriovenous or venovenous haemofiltration provides gentle, continuous dialysis and fluid removal. Acute renal failure in childhood generally carries a good prognosis for renal recovery unless complicating a life-threatening condition, e.g. severe infection, following cardiac surgery or multisystem failure.

Haemolytic uraemic syndrome

Haemolytic uraemic syndrome (HUS) is a triad of acute renal failure, microangiopathic haemolytic anaemia and thrombocytopenia. Typical HUS is secondary to gastrointestinal infection with verocytotoxin-producing *E. coli* O157:H7, acquired through contact with farm animals or eating uncooked beef, or, less often, *Shigella*. It follows a prodrome of bloody diarrhoea. The toxin from these organisms enters the gastrointestinal mucosa and preferentially localises to the endothelial cells of the kidney where it causes intravascular thrombogenesis. Coagulation cascade is activated and clotting is normal (unlike in disseminated intravascular coagulation, DIC). Platelets are consumed in this process and microangiopathic haemolytic anemia results from damage to red blood cells as they circulate through the microcirculation, which is occluded. Other organs such as the brain, pancreas and heart may also be involved.

With early supportive therapy, including dialysis, the typical diarrhoea-associated HUS usually has a good prognosis, although follow-up is necessary as there may be persistent proteinuria and the development of hypertension and declining renal function in subsequent years. In contrast, atypical HUS has no diarrhoeal prodrome, may be familial and frequently relapses. It has a high risk of hypertension and chronic renal failure and has a high mortality. Children with intracerebral involvement or with atypical HUS may be treated with plasma exchange or plasma infusions, but their efficacy is unproven.

Haemolytic uraemic syndrome (HUS) – the triad of:

- **Acute renal failure**
- **Haemolytic anaemia**
- **Thrombocytopenia.**

Summary

Acute renal failure

- Prerenal: commonest cause in children, from hypovolaemia and circulatory failure
- Renal: most often haemolytic uraemic syndrome or multisystem failure
- Postrenal: from urinary obstruction
- Management: treat underlying cause, metabolic abnormalities, dialysis if necessary.

Chronic kidney disease

Chronic renal failure, with GFR <15 ml/min per 1.73 m^2, is much less common in children than in adults, with an incidence of only 10 per million of the child population each year. Congenital and familial causes are more common in childhood than are acquired diseases (Table 18.6).

Table 18.6 Causes of chronic renal failure

Structural malformations	40%
Glomerulonephritis	25%
Hereditary nephropathies	20%
Systemic diseases	10%
Miscellaneous/unknown	5%

Clinical features

Chronic renal failure presents with:

- Anorexia and lethargy
- Polydipsia and polyuria
- Failure to thrive/growth failure
- Bony deformities from renal osteodystrophy (renal rickets)
- Hypertension
- Acute-on-chronic renal failure (precipitated by infection or dehydration)
- Incidental finding of proteinuria
- Unexplained normochromic, normocytic anaemia.

Many children with chronic renal failure have had their renal disease detected before birth by antenatal ultrasound or have previously identified renal disease. Symptoms rarely develop before renal function falls to less than one-third of normal.

Management

The aims of management are to prevent the symptoms and metabolic abnormalities of chronic renal failure, to allow normal growth and development and to preserve residual renal function. The management of these children should be conducted in a specialist paediatric nephrology centre.

Diet

Anorexia and vomiting are common. Improving nutrition using calorie supplements and nasogastric or gastrostomy feeding is often necessary to optimise growth. Protein intake should be sufficient to maintain growth and a normal albumin, whilst preventing the accumulation of toxic metabolic byproducts.

Prevention of renal osteodystrophy

Phosphate retention and hypocalcaemia due to decreased activation of vitamin D leads to secondary hyperparathyroidism, which results in osteitis fibrosa and osteomalacia. Phosphate restriction by decreasing the dietary intake of milk products, calcium carbonate as a phosphate binder, and activated vitamin D supplements help to prevent renal osteodystrophy.

Control of salt and water balance and acidosis

Many children with chronic renal failure caused by congenital structural malformations and renal dysplasia have an obligatory loss of salt and water. They need

salt supplements and free access to water. Treatment with bicarbonate supplements is necessary to prevent acidosis.

Anaemia

Reduced production of erythropoietin and circulation of metabolites that are toxic to the bone marrow result in anaemia. This responds well to the administration of recombinant human erythropoietin.

Hormonal abnormalities

Many hormonal abnormalities occur in chronic renal failure. Most importantly, there is growth hormone resistance with high growth hormone levels but poor growth. Recombinant human growth hormone has been shown to be effective in improving growth for up to 5 years of treatment, but whether it improves final height remains unknown. Many children with chronic renal failure have delayed puberty and a subnormal pubertal growth spurt.

Dialysis and transplantation

It is now possible for all children to enter renal replacement therapy programmes when end-stage renal failure is reached. The optimum management is by renal transplantation. Technically, this is difficult in very small children and a minimum weight, e.g. 10 kg, needs to be reached before transplantation to avoid renal vein thrombosis. Kidneys obtained from living related donors have a higher success rate than deceased donor kidneys, which are matched as far as possible to the recipient's HLA type. Patient survival is high and first-year graft survival is around 97% for living related and 93% for deceased kidneys in the UK. Graft losses from both acute and chronic rejection or recurrent disease mean that the 5-year graft survival is reduced to 91% for living related kidneys and 79% for deceased donor kidney transplants and some children need re-transplantation. Current immunosuppression is mainly with combinations of prednisolone, tacrolimus and azathioprine or mycophenolate mofetil.

Ideally, a child is transplanted before dialysis is required, but if this is not possible, a period of dialysis may be necessary. Peritoneal dialysis, either by cycling overnight using a machine (continuous cycling peritoneal dialysis) or by manual exchanges over 24 h (continuous ambulatory peritoneal dialysis), can be done by the parents at home and is therefore less disruptive to family life and the child's schooling. Haemodialysis is an alternative and is usually done in hospital 3–4 times a week.

Summary

Chronic renal failure

- Causes: congenital (structural malformations and hereditary nephropathies) most common
- Presentation: abnormal antenatal ultrasound, anorexia and lethargy, polydipsia and polyuria, failure to thrive/growth failure, renal rickets, hypertension, proteinuria, anaemia
- Management: diet and nasogastric or gastrostomy feeding, phosphate restriction and activated vitamin D to prevent renal osteodystrophy, salt supplements and free access to water to control salt and water balance, bicarbonate supplements to prevent acidosis, erythropoietin to prevent anaemia, growth hormone and dialysis and transplantation.

Further reading

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American Society of Pediatric Nephrology: http://www.aspneph.com/patient_info_websites.asp.

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19

Genitalia

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Genital disorders in boys

Most disorders of the genitalia in childhood are in boys.

Inguinoscrotal disorders

Embryology

The testis is formed from the urogenital ridge on the posterior abdominal wall close to the developing kidney. Gonadal induction to form a testis is regulated by genes on the Y chromosome. During gestation, the testis migrates down towards the inguinal canal, guided by mesenchymal tissue known as the gubernaculum, probably under the influence of anti-Müllerian hormone (Fig. 19.1a).

Inguinoscrotal descent of the testis requires the release of testosterone from the fetal testis. A tongue of peritoneum, the processus vaginalis, precedes the migrating testis through the inguinal canal. This peritoneal extension normally becomes obliterated after birth, but failure of this process may lead to the development of an inguinal hernia or hydrocele (Fig. 19.1b–d).

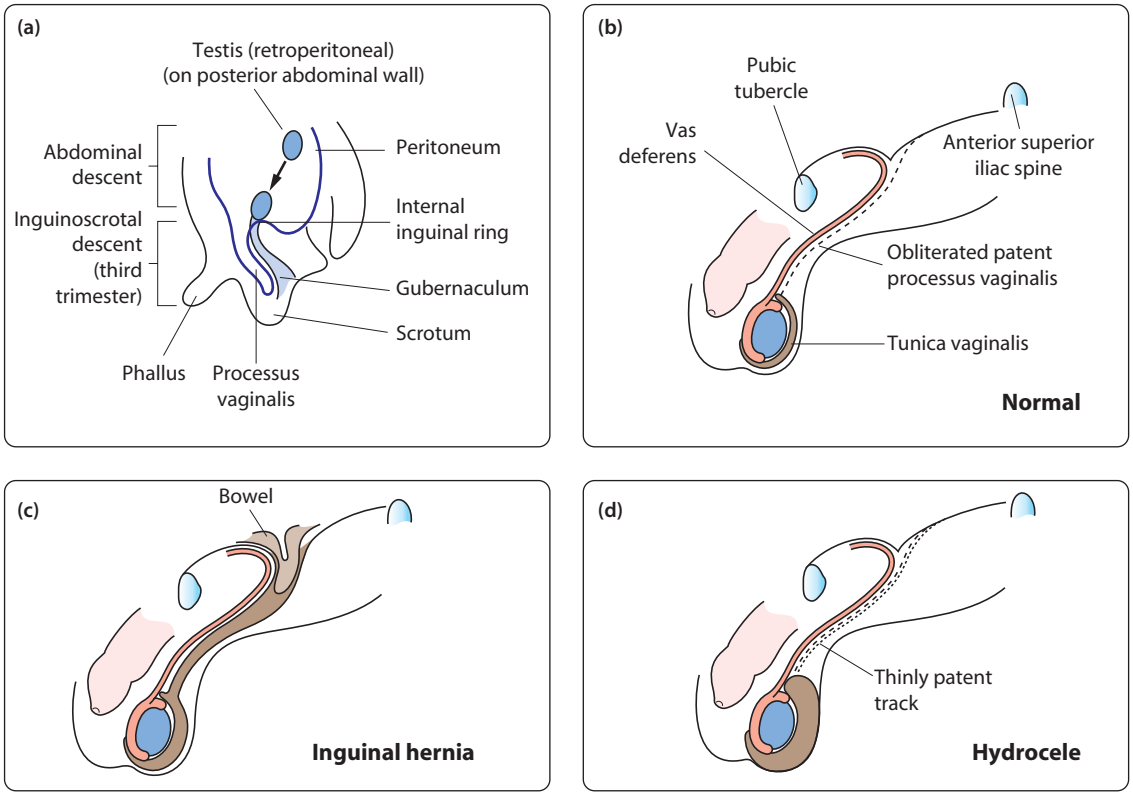


Figure 19.1 **(a)** Embryology of testicular descent. **(b)** The normal testis. **(c)** Inguinal hernia and **(d)** hydrocele are the result of incomplete obliteration of the processus vaginalis.

Inguinal hernia in infants

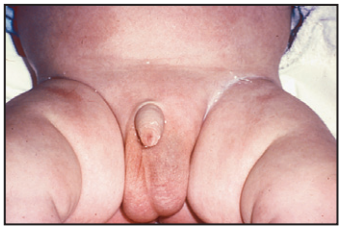


Figure 19.2 Left inguina hernia in an infant. The left groin is only slightly swollen.



Figure 19.3 Bilateral inguina hernias in a preterm infant. Inguinal hernia is primarily a groin swelling; only when it is large does it extend into the scrotum.



Figure 19.4 Right-sided hydrocele. The scrotal swelling often has a bluish discoloration and will transilluminate in a darkened room.

Inguinal hernia

Inguinal hernias in children are almost always indirect and due to a patent processus vaginalis. They are much more frequent in boys and are particularly common in premature infants. Hernias are more common on the right side. At least 1 in 50 boys will develop an inguinal hernia.

Inguinal hernias usually present as an intermittent swelling in the groin or scrotum on crying or straining. Unless the hernia is observed as an inguinal swelling (Figs 19.2, 19.3), diagnosis relies on the history and the identification of thickening of the spermatic cord (or round ligament in girls). The groin swelling may become visible on raising the intra-abdominal pressure by gently pressing on the abdomen or asking the child to cough.

An inguinal hernia in an infant may present as an irreducible lump in the groin or scrotum. The lump is firm and tender. The infant may be unwell with irritability and vomiting. Most 'irreducible' hernias can be successfully reduced following opioid analgesia and sustained gentle compression. Surgery is delayed for 24–48 hours to allow resolution of oedema. If reduction is impossible, emergency surgery is required because of the risk of strangulation of bowel and damage to the testis. A hernia associated with an undescended testis should be operated early to minimise risks to the testis.

Surgery

The operation is carried out via an inguinal skin crease incision and involves ligation and division of the hernial sac (processus vaginalis). Except in small infants, this can usually be undertaken as a day-case procedure, provided there is appropriate anaesthetic and surgical support.

Inguinal hernias in infants should be repaired promptly to avoid the risk of strangulation.

Hydrocele

A patent processus vaginalis, which is sufficiently narrow to prevent the formation of an inguinal hernia, may still allow peritoneal fluid to track down around the testis to form a hydrocele (Fig. 19.4). Hydroceles are asymptomatic scrotal swellings, often bilateral, and sometimes with a bluish discoloration. They may be tense or lax but are non-tender and transilluminate. Some hydroceles are not evident at birth but present in early childhood after a viral or gastrointestinal illness. The majority resolve spontaneously as the processus continues to obliterate, but surgery is considered if it persists beyond 18–24 months of age. A hydrocele of the cord forms a non-tender mobile swelling in the spermatic cord.

Undescended testis

An undescended testis has been arrested along its normal pathway of descent (Fig. 19.5). At birth, about 4% of full-term male infants will have a unilateral or bilateral undescended testis (cryptorchidism). It is more common in preterm infants because testicular descent through the inguinal canal occurs in the third trimester. Testicular descent may continue during early infancy and by 3 months of age the overall rate of cryptorchidism in boys is 1.5%, with little change thereafter. Contrary to previous teaching, it is now recognised that occasionally a testis which is fully descended at birth can *ascend* to an inguinal position during childhood,



Figure 19.5 A left undescended testis with an empty hemiscrotum.

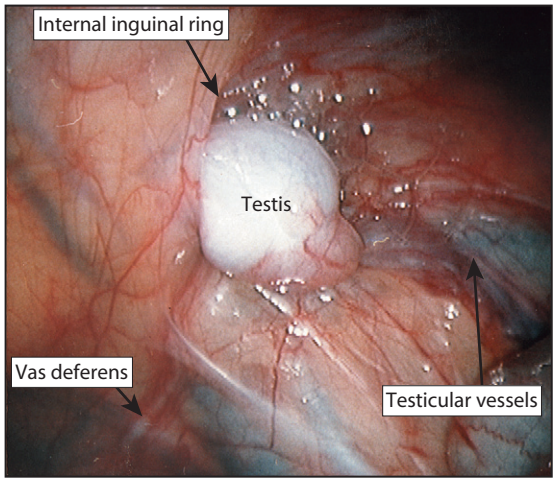


Figure 19.6 Laparoscopic appearance of an intra-abdominal testis.

accounting for some late-presenting 'undescended' or 'ascended' testes. This phenomenon may be due to a relative shortening of cord structures during growth of the child.

Examination

This should be carried out in a warm room, with warm hands and a relaxed child. The testes can then be brought down into a palpable position by gently massaging the contents of the inguinal canal towards the scrotum.

Classification

Retractile

The testis can be manipulated into the bottom of the scrotum without tension, but subsequently retracts into the inguinal region, pulled up by the cremasteric muscle. The testis has usually been found in the scrotum at a neonatal check and been noted by parents on bathing their baby. With age, the testis resides permanently in the scrotum. Follow-up is advisable as, rarely, the testis subsequently ascends into the inguinal canal.

Palpable

The testis can be palpated in the groin but cannot be manipulated into the scrotum. Occasionally, a testis is ectopic, when it lies outside its normal line of descent and may then be found in the perineum or femoral triangle.

Impalpable

No testis can be felt on detailed examination. The testis may be in the inguinal canal, intra-abdominal or absent.

Investigations

Useful investigations include:

- *Ultrasound* – this has a limited role in identifying testes in the inguinal canal in obese boys but cannot reliably distinguish between an intra-abdominal or absent testis. It is performed in children with bilateral impalpable testes to verify internal pelvic organs.
- *Hormonal* – for bilateral impalpable testes, the presence of testicular tissue can be confirmed by recording a rise in serum testosterone in response to intramuscular injections of human chorionic gonadotrophin (HCG); these boys may require specialist endocrine review.
- *Laparoscopy* (Fig. 19.6) – the investigation of choice for the impalpable testis. Under anaesthesia, inguinal examination is first carried out to check that the testis is not in the inguinal canal.

Management

Surgical placement of the testis in the scrotum (*orchidopexy*) is undertaken for several reasons:

- *Fertility* – to optimise spermatogenesis, the testis needs to be in the scrotum below body temperature. The timing of orchidopexy is controversial, but orchidopexy during the second year of life may optimise reproductive potential. After 6 months of age descent of testis is unlikely and referral for paediatric surgical review at that age is recommended. Fertility after orchidopexy for a unilateral undescended testis is close to normal. In contrast, fertility is reduced to around 50% after bilateral orchidopexy for palpable undescended testes, and men with a history of bilaterally impalpable testes are usually sterile.
- *Malignancy* – undescended testes have histological abnormalities and an increased risk of malignancy. The risk is greater for bilateral undescended testes and the greatest risk is for testes which are intra-abdominal. Although the evidence is somewhat contradictory, some studies have suggested that early orchidopexy for a unilateral undescended testis reduces the risk to nearly the same as a normal testis. A scrotal testis can also be more easily self-examined than an inguinal or ectopic one.

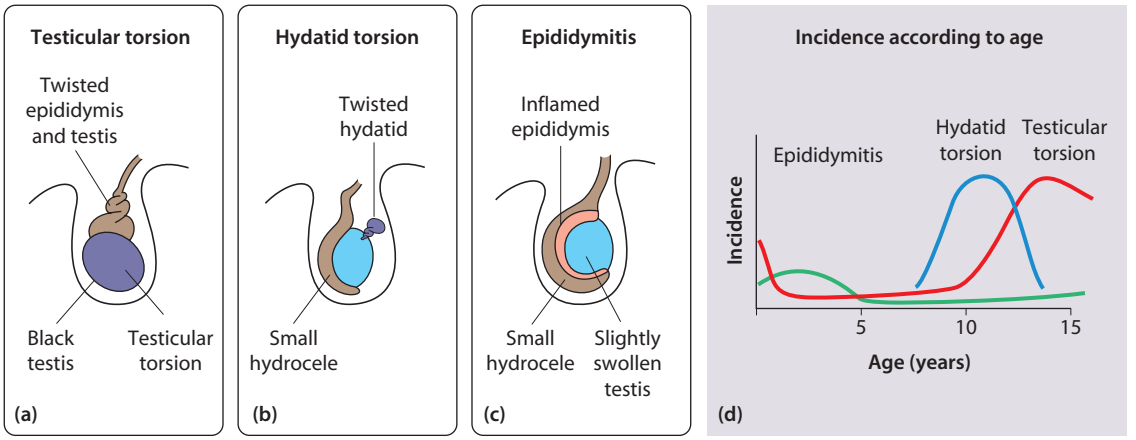


Figure 19.7 **(a)** Testicular torsion. **(b)** Hydatid torsion. **(c)** Epididymitis. **(d)** Incidence in relation to age.

- *Cosmetic and psychological* – if a testis is absent, a prosthesis can be used but this is best delayed until a larger adult-sized prosthesis can be inserted.

Surgery

Most boys with an undescended testis undergo an orchidopexy via an inguinal incision. The testis is mobilised, preserving the vas deferens and testicular vessels, the associated patent processus vaginalis is ligated and divided, and the testis is placed in the scrotal pouch. The operation is usually performed as a day-case procedure.

Most intra-abdominal testes are amenable to laparoscopic staged orchidopexy. Orchidectomy is considered in rare circumstances of a very high testis in an older child or a unilateral intra-abdominal testis in a postpubertal adolescent with a normal contralateral testis. Before conducting laparoscopy in a peripubertal boy, baseline abdominal ultrasound scan and serum tumour markers are checked. Although intra-abdominal testes have profoundly defective spermatogenesis, they are capable of producing male hormones.

Varicocele

Varicosities of the testicular veins may develop in boys around puberty. They are usually on the left side and there is an association with subfertility. Treatment is indicated for symptoms (dragging, aching), impaired testicular growth and, in later life, for infertility. Obliteration of the testicular veins can be achieved by conventional surgery, laparoscopic techniques or radiological embolisation. The role of such interventions in asymptomatic boys is uncertain.

The acute scrotum

Torsion of the testis

Testicular torsion is most common in adolescents but may occur at any age, including the perinatal period (Fig. 19.7). The pain is not always centred on the scrotum but may be in the groin or lower abdomen. Atypical presentation is not unusual and the testes must always be examined whenever a boy or young man presents with inguinal or lower abdominal pain of sudden onset (see Case History 19.1). There may be a history of previous self-limiting episodes. Torsion of the testis must be relieved within 6–12 h of the onset of symptoms for there to be a good chance of testicular viability. Surgical exploration is mandatory unless torsion can be excluded. If torsion is confirmed, fixation of the contralateral testis is essential because there may be an anatomical predisposition to torsion, for example the 'bell clapper' testis, where the testis is not anchored properly. An undescended testis is at increased risk of torsion and at increased risk of delayed diagnosis. It may also be confused with an incarcerated hernia. Expert Doppler ultrasound looking at flow in the testicular blood vessels may allow torsion of the testis to be differentiated from epididymitis, but should not be used to diagnose torsion as only early surgical correction may salvage the testis. If there is any doubt about the cause of a painful scrotum, surgery should be performed.

Torsion of testicular appendage

A hydatid of Morgagni is an embryological remnant found on the upper pole of the testis. Torsion of this appendage characteristically affects boys just prior to puberty. This may be because of rapid enlargement of the hydatid in response to gonadotrophins. The pain may increase over 1 or 2 days and occasionally the torted hydatid can be seen or felt (the blue dot sign). Surgical exploration and excision of the appendage leads to rapid resolution of the problem.

Other causes

Viral or bacterial epididymo-orchitis or epididymitis may cause an acute scrotum in infants and toddlers, and scrotal exploration is often necessary to confirm the diagnosis. If an associated urinary tract infection is present, antibiotic treatment and full investigation of the urinary tract will be required. Other conditions which may cause scrotal symptoms and signs are idiopathic scrotal oedema (usually painless, bilateral

Case History

19.1 Torsion of the testis

A 13-year-old boy presents to the A&E Department with a 2-hour history of right lower abdominal pain of sudden onset. He has vomited once. Temperature 37.4°C. He indicates that his pain is in the right lower quadrant. Urine dipstick testing was normal. Appendicitis is suspected. However, examination of the abdomen does not reveal any guarding or other signs of peritoneal irritation in the right iliac fossa. When his testes are examined, the right testis is found to be slightly swollen and lying higher in the scrotum than the left testis (Fig. 19.8). Although he has not complained of testicular pain, the testis is tender on palpation. Urgent surgical exploration confirms testicular torsion (Fig. 19.9). After detorsion, the testis appears viable and is conserved. It is fixed with sutures to minimise the risk of further torsion. The left testis is also fixed, as the anatomical variant which predisposes to torsion occurs bilaterally.

This case highlights:

- The clinical features of testicular torsion are variable and can be potentially misleading, with pain predominantly referred to the abdomen or inguinal region and minimal pain felt in the testis itself
- Abdominal examination is never complete without inspection and gentle palpation of both testes
- With torsion, the testis is always tender.



Figure 19.8 Enlarged, raised right testis, which was tender on palpation.



Figure 19.9 Torsion of the testis at surgery.

scrotal swelling and redness in a preschool child) or an incarcerated inguinal hernia.

Torsion of the testis is an emergency.

Abnormalities of the penis

Hypospadias

In the male fetus, urethral tubularisation occurs in a proximal to distal direction under the influence of fetal testosterone. Failure to complete this process leaves the urethral opening proximal to the normal meatus on the glans and this is termed hypospadias (Fig. 19.10). This is a common congenital anomaly, affecting about 1 in every 200 boys. Recent studies suggest that the incidence is increasing.

Hypospadias consists of:

- *A ventral urethral meatus* – in most cases the urethra opens on or adjacent to the glans penis, but in severe cases the opening may be on the penile shaft or in the perineum (Fig. 19.11)
- *A hooded dorsal foreskin* – the foreskin has failed to fuse ventrally
- *Chordee* – a ventral curvature of the shaft of the penis, most apparent on erection. This is only marked in the more severe forms of hypospadias (Fig. 19.12).

Glanular hypospadias may be a solely cosmetic concern, but more proximal varieties may cause functional problems including an inability to micturate in a normal direction and erectile deformity. With more severe varieties of hypospadias, additional genitourinary anomalies should be excluded and sometimes it is necessary to consider disorders of sexual differentiation.

Surgery

Correction is often undertaken before 2 years of age, often as a single-stage operation. The aims of surgery are to produce:

- A terminal urethral meatus so that the boy can micturate in a normal standing position like his peers
- A straight erection
- A penis that looks normal.

Infants with hypospadias must not be circumcised, as the foreskin is often needed for later reconstructive surgery.

Hypospadias

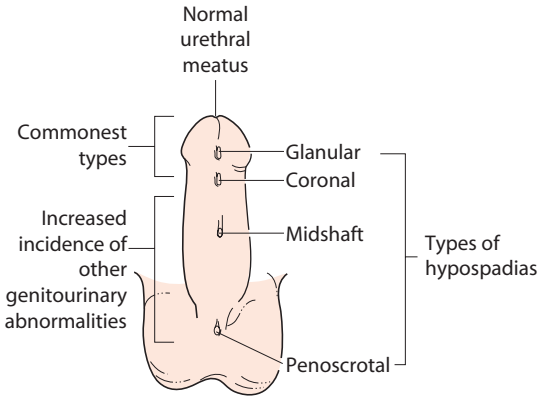


Figure 19.10 Varieties of hypospadias.

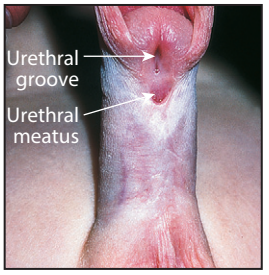


Figure 19.11 Penile shaft hypospadias with dorsal hooded foreskin, showing the urethral groove (arrow) and urethral meatus (arrow).



Figure 19.12 In lateral view, the ventral curvature of the penis (chordee) can be seen.

Circumcision

At birth, the foreskin is adherent to the surface of the glans penis. These adhesions separate spontaneously with time, allowing the foreskin to become more mobile and eventually retractile. At 1 year of age, approximately 50% of boys have a *non-retractile foreskin*, but by 4 years this has declined to 10%, and by 16 years to only 1%. A non-retractile foreskin often leads to ballooning on micturition, which is physiological. Gentle retraction of the foreskin at bathtimes helps to maintain hygiene, but forcible retraction of a healthy non-retractile foreskin should be avoided.

Two conditions that require reassurance are preputial adhesions (when the foreskin remains partially adherent to the glans) and the presence of white 'pearls' under the foreskin due to trapped epithelial squames. Both conditions are usually asymptomatic and resolve spontaneously.

Circumcision is one of the earliest recorded operations and remains an important tradition in the Jewish and Muslim religions. Although routine neonatal circumcision is still common in some Western countries such as the USA, the arguments generally used to justify on medical grounds have been discredited and no national or international medical association currently advocates routine neonatal circumcision. Neonatal circumcision is not without risk of significant morbidity. Nevertheless, the issue is still hotly debated.

There are only a few medical indications for circumcision:

- *Phimosis* (Fig. 19.13). This term is often used to describe the inability to retract the foreskin. As described above, at birth the foreskin is non-retractile and phimosis is physiological. Pathological phimosis is seen as a whitish scarring of the foreskin and is rare before the age of 5 years. The condition is due to a localised skin disease known as balanitis xerotica obliterans (BXO), which also involves the glans penis and can cause urethral meatal stenosis



Figure 19.13 Pathological phimosis.



Figure 19.14 Balanoposthitis.

- *Recurrent balanoposthitis* (Fig. 19.14). Single attack of redness and inflammation of the foreskin, sometimes with a purulent discharge, is common and usually responds rapidly to warm baths and a broad-spectrum antibiotic. Recurrent attacks of balanoposthitis (inflammation of the glans and foreskin) are uncommon and circumcision is occasionally indicated.
- *Recurrent urinary tract infections*. Although urinary infection is more common in uncircumcised boys

the overall incidence is low and routine circumcision is not justified as a preventative measure. However, circumcision may be helpful in reducing the risk of urinary tract bacterial colonisation in boys with upper urinary tract anomalies complicated by recurrent urinary infection. It may also be appropriate in boys with spina bifida who need to perform clean intermittent urethral catheterisation.

There is some data from countries with a high prevalence of HIV infection that the risk of transmission is lower in circumcised males.

Surgery

Circumcision for medical indications is performed under a general anaesthetic as a day case. During the procedure, a long-acting local anaesthetic block can be given to reduce postoperative pain. Circumcision is not a trivial operation. Healing can take up to 10 days, with discomfort for several days. Bleeding and infection are well-recognised complications, but more serious hazards, such as damage to the glans, may occur if the procedure is not carried out by appropriately trained personnel. The procedure also carries the risk of psychological trauma.

Preputioplasty can be offered as an effective alternative to circumcision in selected cases. After retraction of the foreskin, the tight preputial ring is incised longitudinally and then sutured transversely. Unlike circumcision, preputioplasty conserves the foreskin and results in less postoperative discomfort and fewer complications. However, regular retraction of the foreskin is required in the first few weeks after surgery and for this reason, preputioplasty is better suited to older boys who are willing to do this.

Topical corticosteroids

Application of a topical steroid ointment to the prepuce has been shown to facilitate retraction of a non-retractile prepuce, with success rates of up to 80%. Different treatment regimens have been described but typically the ointment is applied twice daily for 2–3 months.

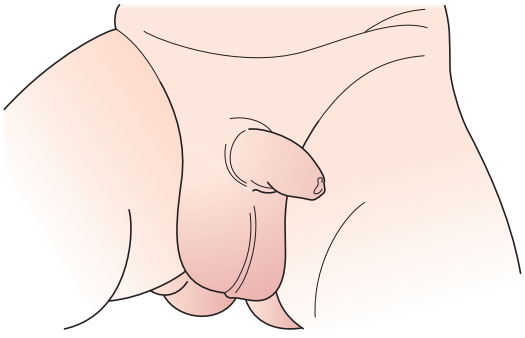
Paraphimosis

The foreskin becomes trapped in the retracted position proximal to a swollen glans. The foreskin can usually be reduced, but adequate analgesia (often a general anaesthetic) is needed to achieve this. The problem is not usually recurrent and circumcision is rarely required.

Summary

Genital conditions in male infants and children

Inguinal hernia:
- Presentation – intermittent swelling in the groin or scrotum on crying or as an irreducible lump
- Repair promptly to avoid the risk of strangulation
- If irreducible – sustained gentle compression with analgesia to reduce, followed by delayed surgery



Hypospadias:
- Consists of – ventral urethral meatus, a hooded dorsal foreskin, chordee
- When severe, exclude other genitourinary or endocrine anomalies
- Affected infants must not be circumcised, as the foreskin is often needed for later reconstructive surgery

An undescended testis:
- Is present in about 4% of full-term male infants but only 1.3% at 3 months of age
- May be **retractile**, if it can be brought to bottom of scrotum without tension but subsequently retracts – is usually normal
- Is **palpable** if felt in the groin but cannot be manipulated into the scrotum
- Is **impalpable** if no testis can be felt – may be in the inguinal canal, intra-abdominal or absent
- Orchidopexy – performed to optimise fertility, avoid malignant change, and for cosmetic and psychological reasons

Torsion of the testis:
- Must always be considered in a boy with an acutely painful scrotum
- Must be treated within hours for the testis to be viable

Circumcision:
- Is not recommended routinely, but is a tradition for Jews and Muslims and is still common in the USA
- The only medical indications are – pathological phimosis, recurrent balanoposthitis and possibly some boys with recurrent urinary tract infections
- Complications include pain, bleeding, infection and damage to the glans

Genital disorders in girls

Inguinal hernias

These are much less common than in boys. Sometimes the ovary becomes incarcerated in the hernial sac and can be difficult to reduce. Rarely, androgen insensitivity syndrome (testicular feminisation) can present as a hernia in a phenotypic female who actually has a male genotype.

Labial adhesions

If the labia minora are adherent in the midline, this may give the appearance of absence of the vagina, except there is a characteristic translucent midline raphe partially or totally occluding the vaginal opening. Asymptomatic adhesions can be left alone and will often lyse spontaneously. If there is perineal soreness or urinary irritation, treatment with topical oestrogen treatment applied sparingly twice a day for 1–2 weeks often dissolves the adhesions. Active separation of the adhesions under anaesthesia is sometimes required.

Vulvovaginitis/vaginal discharge

Vulvovaginitis and vaginal discharge are common in young girls. They may result from infection (bacterial or fungal), specific irritants, poor hygiene or sexual abuse, although none of these factors is present in most cases. Vulvovaginitis may rarely be associated with threadworm infestation. Parents should be advised about hygiene, the avoidance of bubble bath and scented soaps and the use of loose-fitting cotton underwear. Swabs should be taken to identify any pathogens, which can then be specifically treated. Salt baths may be helpful. Oestrogen cream applied sparingly to the vulva may relieve the problem in resistant cases by increasing vaginal resistance to infection as prepubertal tissues tend to be atrophic. If there are any concerns about sexual abuse, the child must be seen by a paediatrician (see Ch. 7). Rarely, if the vaginal discharge is persistent or purulent, examination under anaesthesia may be needed to exclude a vaginal foreign body or unusual infections.

Disorders of sexual differentiation are considered in Chapter 11.

Further reading

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20

Liver disorders

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Features of liver disorders in children are:

- Prolonged (persistent) neonatal jaundice is the most common presentation of liver disease in the neonatal period.
- The earlier in life biliary atresia is diagnosed and treated surgically, the better the prognosis.
- Transmission of hepatitis B surface antigen-positive mothers is prevented by immunising their babies.
- Chronic liver disease (Fig. 20.1), cirrhosis and portal hypertension are uncommon and should be managed by, or in conjunction with tertiary or national centres.
- Liver transplantation is an effective therapy for acute or chronic liver failure, with a >80% 5-year survival rate; it is only performed at national centres.

Neonatal liver disease

Many newborn infants become clinically jaundiced. About 5–10% are still jaundiced at >2 weeks of age (3 weeks if preterm), when it is called 'prolonged (or persistent) neonatal jaundice'. This is usually an unconjugated hyperbilirubinaemia, which resolves spontaneously (Box 20.1). Prolonged neonatal jaundice caused by liver disease is characterised by a raised conjugated bilirubin (>20 µmol/L) and is usually accompanied by:

- Pale stools
- Dark urine
- Bleeding tendency
- Failure to thrive.

There is an urgency to diagnose liver disease as early as possible in the neonatal period, because early diagnosis and management improves prognosis.

In prolonged (persistent) neonatal jaundice, check if it is conjugated hyperbilirubinaemia, as this is due to liver disease.

Bile duct obstruction

Biliary atresia (see Case History 20.1)

This occurs in 1 in 14 000 live births. It is a progressive disease, in which there is destruction or absence of the extrahepatic biliary tree and intrahepatic biliary ducts.

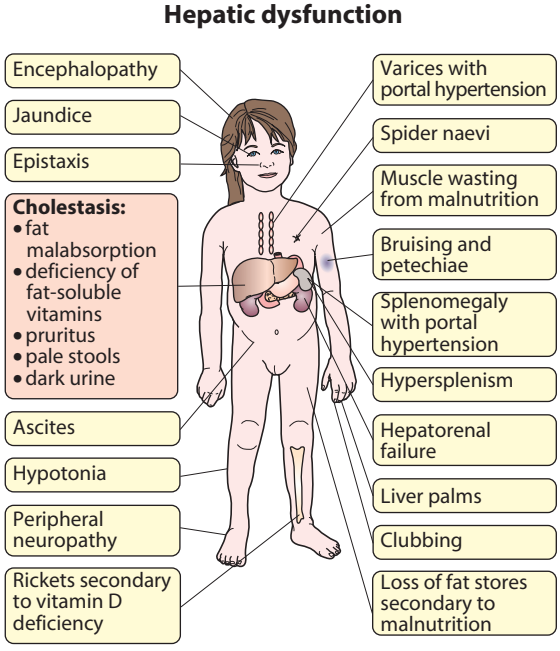


Figure 20.1 Clinical features of liver disease. In addition, these children may have growth failure and developmental delay.

Box 20.1 Causes of prolonged (persistent) neonatal jaundice

Unconjugated

- Breast milk jaundice
- Infection (particularly urinary tract)
- Haemolytic anaemia, e.g. G6PD deficiency
- Hypothyroidism
- High gastrointestinal obstruction
- Crigler–Najjar syndrome

Conjugated (>20 micromol/L)

Bile duct obstruction

- Biliary atresia
- Choledochal cyst

Neonatal hepatitis syndrome

- Congenital infection
- Inborn errors of metabolism
- α_1-Antitrypsin deficiency
- Galactosaemia
- Tyrosinaemia (type 1)
- Errors of bile acid synthesis
- Progressive familial intrahepatic cholestasis (PFIC)
- Cystic fibrosis
- Intestinal failure-associated liver disease – associated with long-term parenteral nutrition

Intrahepatic biliary hypoplasia

- Alagille syndrome.

This leads to chronic liver failure and death unless surgical intervention is performed. Babies with biliary atresia have a normal birthweight but fail to thrive as the disease progresses. They are usually mildly jaundiced and, following passage of meconium, their stools are pale and their urine dark. Although stool colour may fluctuate, pale stools is an important abnormality and warrants investigation, even in the absence of clinical jaundice. Hepatomegaly is often present and splenomegaly will develop secondary to portal hypertension.

Standard liver function tests are of little value in the differential diagnosis. A fasting abdominal ultrasound may demonstrate a contracted or absent gallbladder, though it may be normal. A radioisotope scan with TIBIDA (iminodiacetic acid derivatives) shows good uptake by the liver, but no excretion into the bowel. Liver biopsy demonstrates features of extrahepatic biliary obstruction, although features may overlap with those of neonatal hepatitis, especially if carried out at an early stage of the disease. The diagnosis is confirmed at laparotomy by operative cholangiography which fails to outline a normal biliary tree.

Treatment consists of surgical bypass of the fibrotic ducts, hepatoportoenterostomy (Kasai procedure), in which a loop of jejunum is anastomosed to the cut surface of the porta hepatis, facilitating drainage of bile from any remaining patent ductules. If surgery is performed before the age of 60 days, 80% of children achieve bile drainage. The success rate diminishes with increasing age – hence the need for early diagnosis and treatment. Postoperative complications include cholangitis and malabsorption of fats and fat-soluble vitamins. Even when bile drainage is successful, there is frequently progression to cirrhosis and portal hypertension. If the operation is unsuccessful, liver transplantation has to be considered. Biliary atresia is the single most common indication for liver transplantation in the paediatric age group.

Choledochal cysts

These are cystic dilatations of the extrahepatic biliary system. About 25% present in infancy with cholestasis. In the older age group, choledochal cysts present with abdominal pain, a palpable mass and jaundice or cholangitis. The diagnosis is established by ultrasound or radionuclide scanning. Treatment is by surgical excision of the cyst with the formation of a Roux-en-Y anastomosis to the biliary duct. Future complications include cholangitis and a 2% risk of malignancy, which may develop in any part of the biliary tree.

Neonatal hepatitis syndrome

In neonatal hepatitis syndrome, there is prolonged neonatal jaundice and hepatic inflammation. Its causes are listed in Box 20.1, but often no specific cause is identified. In contrast to biliary atresia, these infants may have intrauterine growth restriction and hepatosplenomegaly at birth. Liver biopsy (Fig. 20.6) is often non-specific, demonstrating a giant cell hepatitis.

α_1-Antitrypsin deficiency

Deficiency of the protease α_1-antitrypsin is associated with liver disease in infancy and childhood and emphysema in adults. It is inherited as an autosomal recessive disorder with an incidence of 1 in 2000–4000 in the UK. There are many phenotypes of the protease inhibitor (Pi) which are coded on chromosome 14. Liver disease is primarily associated with the phenotype PiZZ.

The majority of children who present with α_1-antitrypsin deficiency will either have prolonged neonatal jaundice or, less commonly, bleeding due to vitamin K deficiency (haemorrhagic disease of the newborn). Hepatomegaly is present. Splenomegaly develops with cirrhosis and portal hypertension. The diagnosis is confirmed by estimating the level of α_1-antitrypsin in the plasma and identifying the phenotype. Approximately

Case History

20.1 Biliary atresia

A term infant was given oral vitamin K shortly after birth. He was breast-fed. He became mildly jaundiced on the third day of life. At 5 weeks of age, he presented with poor feeding and vomiting and a history of bruising on his forehead and shoulders. His urine had become dark and stools intermittently pale. He was pale, jaundiced, had several bruises and hepatomegaly. Investigations showed:

- Hb 8.8 g/L
- Platelets 465×10^9/L
- Prothrombin time – 28 s (normal range 10–13 s)
- Serum bilirubin 178 micromol/L – 140 micromol/L conjugated.

The investigation of conjugated hyperbilirubinaemia is shown in Figure 20.2. The TIBIDA radionuclide scan showed no excretion at 24 h (Fig. 20.3) and a liver biopsy suggested biliary atresia (Fig. 20.4). A hepatoportoenterostomy was performed at 6 weeks of age (Fig. 20.5).

In persistent neonatal jaundice, early diagnosis of biliary atresia improves the prognosis.

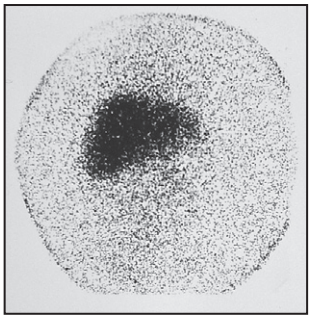


Figure 20.3 Radioisotope scan TIBIDA of liver showing good hepatic uptake of isotope and no excretion into bowel. This scan suggests extrahepatic biliary obstruction or atresia or severe intrahepatic cholestasis.

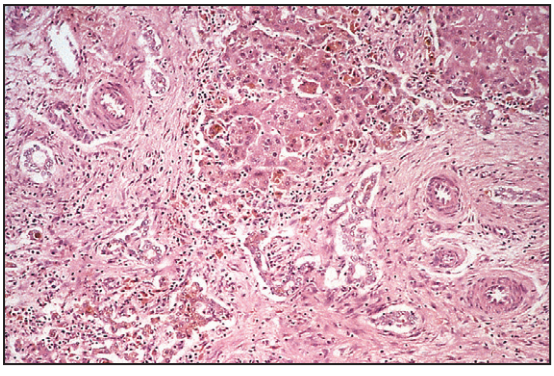


Figure 20.4 Liver biopsy of biliary atresia showing bands of fibrous tissue with bile duct proliferation.



Figure 20.5 Shortly after successful bile drainage by hepatoportoenterostomy (Kasai procedure) for biliary atresia.

Evaluation of neonatal conjugated hyperbilirubinaemia

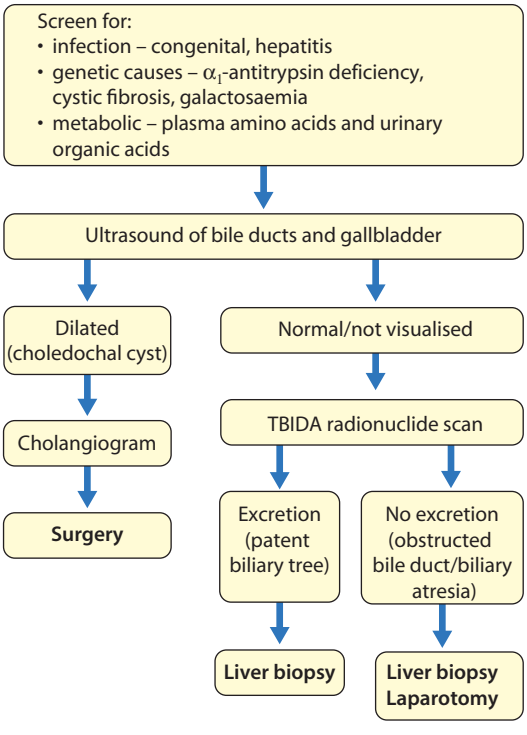


Figure 20.2 Evaluation of neonatal conjugated hyperbilirubinaemia.

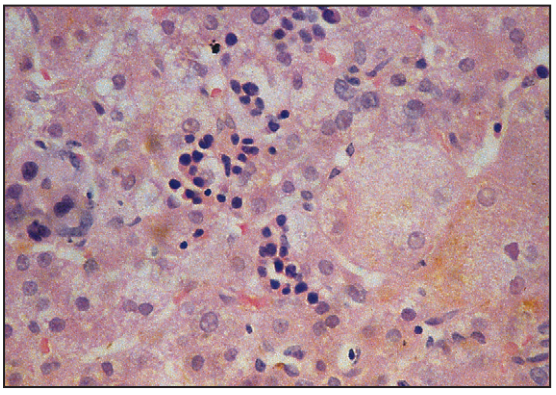


Figure 20.6 Liver biopsy in neonatal hepatitis showing inflammatory infiltrate throughout the liver, and giant cell and rosette formation of liver cells.

50% of children have a good prognosis, but the remainder will develop liver disease and may require transplantation. Pulmonary disease is not significant in childhood, but is likely to develop in adult life. Advice to avoid smoking (both active and passive) should be given. The disorder can be diagnosed antenatally.

Galactosaemia

This very rare disorder has an incidence of 1 in 40 000. The infants develop poor feeding, vomiting, jaundice and hepatomegaly when fed milk. Liver failure, cataracts and developmental delay are inevitable if galactosaemia is untreated. A rapidly fatal course with shock, haemorrhage and disseminated intravascular coagulation, often due to Gram-negative sepsis, may occur.

The condition can be screened for, in prolonged (persistent) jaundice, by detecting galactose, a reducing substance, in the urine. The diagnosis is made by measuring the enzyme galactose-1-phosphate-uridyl transferase in red cells. A recent blood transfusion may mask the diagnosis. A galactose-free diet prevents progression of liver disease, but ovarian failure and learning difficulties may occur later.

Other causes

Neonatal hepatitis may occur following prolonged parenteral nutrition. Rare causes include tyrosinaemia type 1, cystic fibrosis, lipid and glycogen storage disorders, peroxisomal disorders, inborn errors of bile acid synthesis and progressive familial intrahepatic cholestasis (PFIC).

Inborn errors of bile acid synthesis

Patients presenting with neonatal cholestasis of infancy and normal levels of gamma glutamyl transferase (GGT) should be screened for elevated cholenoic bile acids in urine. Diagnosis is confirmed by mass spectrometry of urine for bile acids. Specific treatment is with ursodeoxycholic acid.

Progressive familial intrahepatic cholestasis (PFIC)

This is a heterogeneous group of cholestatic disorders of bile transporter defects caused by recessive mutations in different genes. Children present with jaundice, intense pruritus, diarrhoea with failure to thrive, rickets and a variable progression of liver disease. In two forms of the disorder, the gamma glutamyl transferase (GGT) is low. Prognosis is variable; some children require liver transplantation.

Intrahepatic biliary hypoplasia

Syndromic causes

Alagille syndrome is a rare autosomal dominant condition with widely varying penetrance. Infants may have characteristic triangular facies, skeletal abnormalities, congenital heart disease (classically peripheral pulmonary stenosis), renal tubular disorders, defects in the eye and intrahepatic biliary hypoplasia with severe pruritus and failure to thrive. Prognosis is variable, with 50% of children surviving into adult life without liver transplantation. Intrahepatic biliary hypoplasia also occurs in Down syndrome; there is also a non-syndromic biliary hypoplasia.

In persistent jaundice, always look to see if the stools are pale – suggests bile duct obstruction.

Viral hepatitis

The clinical features of viral hepatitis include nausea, vomiting, abdominal pain, lethargy and jaundice; however, 30–50% of children do not develop jaundice. A large tender liver is common and 30% will have splenomegaly. The liver transaminases are usually markedly elevated. Coagulation is usually normal.

Hepatitis A

Hepatitis A virus (HAV) is an RNA virus which is spread by faecal–oral transmission. The incidence of hepatitis A in childhood has fallen as socioeconomic conditions have improved. Many adults are not immune. Vaccination is required for travellers to endemic areas.

The disease may be asymptomatic, but the majority of children have a mild illness and recover both clinically and biochemically within 2–4 weeks. Some may develop prolonged cholestatic hepatitis (which is self-limiting), or fulminant hepatitis. Chronic liver disease does not occur.

The diagnosis can be confirmed by detecting IgM antibody to the virus.

There is no treatment and no evidence that bed rest or change of diet is effective. Close contacts should be given prophylaxis with human normal immunoglobulin (HNIG) or vaccinated within 2 weeks of the onset of the illness.

Hepatitis B

Hepatitis B virus (HBV) is a DNA virus which is an important cause of acute and chronic liver disease worldwide, with high prevalence and carrier rates in the Far

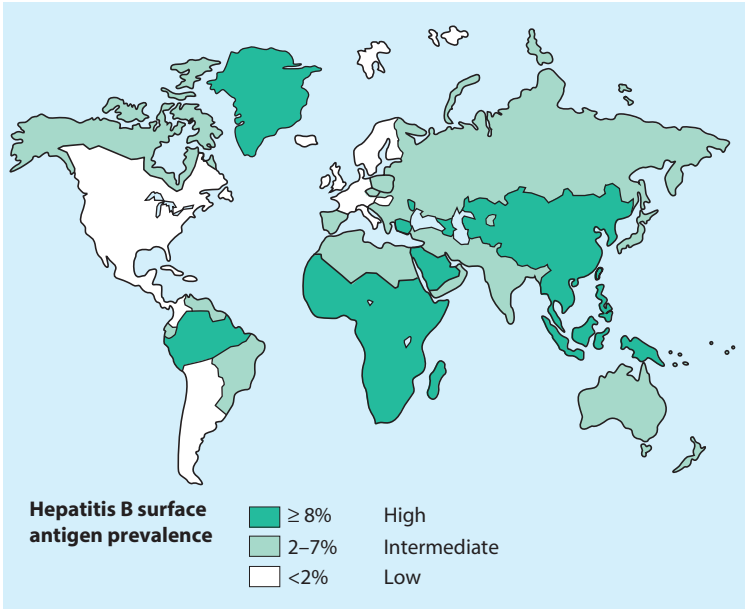


Figure 20.7 Worldwide prevalence of hepatitis B (HBsAg), showing high prevalence in the Far East, sub-Saharan Africa and parts of North and South America (CDC, Centers for Disease Control and Prevention, 2002).

East, sub-Saharan Africa and parts of North and South America (Fig. 20.7). HBV is transmitted by:

- Perinatal transmission from carrier mothers
- Transfusion of infected blood or blood products
- Needlestick injuries with infected blood
- Renal dialysis
- Horizontal spread within families.

Among adults it can also be transmitted sexually.

Infants who contract HBV perinatally are asymptomatic, but at least 90% become chronic carriers. Older children who contract HBV may be asymptomatic or have classical features of acute hepatitis. The majority will resolve spontaneously, but 1–2% develop fulminant hepatic failure, while 5–10% become chronic carriers. The diagnosis is made by detecting HBV antigens and antibodies. IgM antibodies to the core antigen (anti-HBc) are positive in acute infection. Positivity to hepatitis B surface antigen (HBsAg) denotes ongoing infectivity. There is no treatment for acute HBV infection.

Chronic hepatitis B

Infants infected with HBV by vertical transmission from their mothers usually become asymptomatic carriers. Approximately 30–50% of carrier children will develop chronic HBV liver disease, which may progress to cirrhosis in 10%. There is a long-term risk of hepatocellular carcinoma. Current treatment regimens for chronic HBV have poor efficacy. Interferon treatment for chronic hepatitis B is successful in 50% of children infected horizontally and 30% of children infected perinatally. Oral antiviral therapy such as lamivudine is effective in 23% but is limited by the development of resistance. Newer drugs such as adefovir or long-acting (peg) interferon may be more effective.

Prevention

Prevention of HBV infection is important. All pregnant women should have antenatal screening for HBsAg. Babies of all HBsAg-positive mothers should receive a course of hepatitis B vaccination (given routinely in many countries), with hepatitis B immunoglobulin also being given if the mother is also hepatitis B e antigen (HBeAg)-positive. Antibody response to the vaccination course should be checked in high-risk infants as 5% require further vaccination. Other members of the family should also be vaccinated. There is evidence that effective neonatal vaccination reduces the incidence of HBV-related cancer.

Summary

Hepatitis B virus (HBV)

- Perinatal transmission from carrier mothers should be prevented by maternal screening and giving the infant a course of hepatitis B vaccine with hepatitis B immunoglobulin if indicated
- Infection may result in chronic HBV liver disease, which may progress to cirrhosis and hepatocellular carcinoma.

Hepatitis C

Hepatitis C virus (HCV) is an RNA virus that was responsible for 90% of post-transfusion hepatitis until the screening of donor blood was introduced in 1991. In the UK, about 1 in 2000 donors have HCV antibodies. The prevalence is high among intravenous drug users. Children previously at risk were those who received unscreened blood or blood products, in particular

those with haemoglobinopathies or haemophilia. Vertical transmission occurs in 6% from infected mothers but is twice as common if there is co-infection with HIV, and is now the commonest cause of HCV transmission in children. It seldom causes an acute infection, but the majority become chronic carriers, with a 20–25% lifetime risk of progression to cirrhosis or hepatocellular carcinoma. Treatment is with a combination of pegylated interferon and ribavirin. Success of treatment is dependent on virus genotype. In children with genotypes 2/3, it is effective in up to 90% of cases. Treatment is not undertaken before 4 years of age, as it may resolve spontaneously following vertically acquired infections.

Hepatitis D virus

Hepatitis D virus (HDV) is a defective RNA virus that depends on hepatitis B virus for replication. It occurs as a co-infection with hepatitis B virus or as a superinfection causing an acute exacerbation of chronic hepatitis B virus infection. Cirrhosis develops in 50–70% of those who develop chronic HDV infection.

Hepatitis E virus

This is an RNA virus that is enterally transmitted, usually by contaminated water. Epidemics occur in some developing countries.

Non-A to G hepatitis

Clinical presentation is similar to hepatitis A. When a viral aetiology of hepatitis is suspected but not identified, it is known as non-A to G hepatitis.

Epstein–Barr virus

Children with Epstein–Barr virus (EBV) infection are usually asymptomatic. Some 40% have hepatitis that may become fulminant. Less than 5% are jaundiced.

Acute liver failure (fulminant hepatitis)

Acute liver failure in children is the development of massive hepatic necrosis with subsequent loss of liver function, with or without hepatic encephalopathy. The disease is uncommon, but has a high mortality. Most of the cases in childhood are attributed to paracetamol overdosage, non-A to G viral hepatitis and metabolic conditions (Table 20.1). The child may present within hours or weeks with jaundice, encephalopathy, coagulopathy, hypoglycaemia and electrolyte disturbance. Early signs of encephalopathy include alternate periods of irritability and confusion with drowsiness. Older children may be aggressive and unusually difficult. Complications include cerebral oedema, haemorrhage from gastritis or coagulopathy, sepsis and pancreatitis.

Table 20.1 Causes of acute liver failure in children

Infection	Viral hepatitis A, B, C, non-A to G
Poisons/drugs	Paracetamol, isoniazid, halothane, *Amanita phalloides* (poisonous mushroom)
Metabolic	Wilson disease, tyrosinaemia
Autoimmune hepatitis	
Reye syndrome	

Diagnosis

Bilirubin may be normal in the early stages, particularly with metabolic disease. Transaminases are greatly elevated (10–100 times normal), alkaline phosphatase is increased, coagulation is very abnormal and plasma ammonia is elevated. It is essential to monitor the acid–base balance, blood glucose and coagulation times. An EEG will show acute hepatic encephalopathy and a CT scan may demonstrate cerebral oedema.

Management

This includes:

- Maintaining the blood glucose (>4 mmol/L) with intravenous dextrose
- Preventing sepsis with broad-spectrum antibiotics and antifungals
- Preventing haemorrhage, particularly from the gastrointestinal tract with intravenous vitamin K, fresh frozen plasma or cryoprecipitate and H_2-blocking drugs or proton pump inhibitors (PPIs)
- Treating cerebral oedema by fluid restriction and mannitol diuresis
- Urgent transfer to a specialist liver unit.

Features suggestive of a poor prognosis are a shrinking liver, rising bilirubin with falling transaminases, a worsening coagulopathy or progression to coma. Without liver transplantation, 70% of children who progress to coma will die.

Reye syndrome and Reye-like syndrome

Reye syndrome is an acute non-inflammatory encephalopathy with microvesicular fatty infiltration of the liver. Although the aetiology is unknown, there is a close association with aspirin therapy. Since stopping giving aspirin to children aged under 12 years, Reye syndrome has virtually disappeared. With the introduction of tandem mass spectroscopy in the neonatal screening programme, the commonest beta oxidation defect, medium-chain acyl-CoA dehydrogenase (MCAD) deficiency, is diagnosed early in many regions of the UK.

Many of these patients would have presented with a Reye-like syndrome later in life.

Chronic liver disease

The causes of chronic liver disease are given in Box 20.2. The clinical presentation varies from an apparent acute hepatitis to the insidious development of hepatosplenomegaly, cirrhosis and portal hypertension with lethargy and malnutrition. The commonest causes of chronic hepatitis are hepatitis viruses (B or C) and autoimmune hepatitis, but Wilson disease should always be excluded. Histology may demonstrate varying degrees of hepatitis, with an inflammatory infiltrate in the portal tracts that spreads into the liver lobules.

Autoimmune hepatitis

The mean age of presentation is 7–10 years. It is more common in girls. It may present as an acute hepatitis, as fulminant hepatic failure or chronic liver disease with autoimmune features such as skin rash, lupus erythematosus, arthritis, haemolytic anaemia or nephritis. Diagnosis is based on elevated total protein, hypergammaglobulinaemia (IgG >20 g/L); positive autoantibodies, a low serum complement (C4); and typical histology. Autoimmune hepatitis may occur in isolation or in association with inflammatory bowel disease, coeliac disease or other autoimmune diseases. Some 90% of children will respond to prednisolone and azathioprine.

Cystic fibrosis

Liver disease is the second commonest cause of death after respiratory disease in cystic fibrosis. The most common liver abnormality is hepatic steatosis (fatty liver). It may be associated with protein energy malnutrition or micronutrient deficiencies. Steatosis does not generally progress and treatment involves ensuring optimal nutritional support. More significant liver disease arises from thick tenacious bile with abnormal bile acid concentration leading to progressive biliary fibrosis. Cirrhosis and portal hypertension develop in 20% of children by mid-adolescence. Early liver disease is difficult to detect by biochemistry, ultrasound or radioisotope scanning. Liver histology includes fatty liver, focal biliary fibrosis or focal nodular cirrhosis. Therapy includes standard supportive and nutritional therapy and treatment with ursodeoxycholic acid. Liver transplantation may be considered for those with end-stage liver disease, either alone or in combination with a heart–lung transplant.

Wilson disease

Wilson disease is an autosomal recessive disorder with an incidence of 1 in 200 000. Many mutations have now been identified (on chromosome 13). The basic genetic defect is a combination of reduced synthesis of caeruloplasmin (the copper-binding protein) and defective excretion of copper in the bile, which leads to an accumulation of copper in the liver, brain, kidney and cornea. Wilson disease rarely presents in children under the age of 3 years. In those presenting in childhood, a hepatic presentation is more likely. They may present with almost any form of liver disease, including acute hepatitis, fulminant hepatitis, cirrhosis and portal hypertension. Neuropsychiatric features are more common in those presenting from the second decade onwards and include deterioration in school performance, mood and behaviour change, and extrapyramidal signs such as incoordination, tremor and dysarthria. Renal tubular dysfunction, with vitamin D-resistant rickets, and haemolytic anaemia also occur. Copper accumulation in the cornea (Kayser–Fleischer rings) (Fig. 20.8) is not seen before 7 years of age.

Diagnosis can be problematic. A low serum caeruloplasmin and copper is characteristic, but not universal. Urinary copper excretion is increased and this further increases after administering the chelating agent penicillamine. However, the diagnosis is confirmed by the finding of elevated hepatic copper on liver biopsy or identification of the gene mutation.

Treatment is with penicillamine or trientine. Both promote urinary copper excretion, reducing hepatic and central nervous system copper. Zinc is given to

Box 20.2 Causes of chronic liver disease in children

Chronic hepatitis
- Post-viral hepatitis B, C
- Autoimmune hepatitis
- Drugs (nitrofurantoin, non-steroidal anti-inflammatory)
- Inflammatory bowel disease
- Primary sclerosing cholangitis (± ulcerative colitis)

Wilson disease (>3 years)

α_1-Antitrypsin deficiency

Cystic fibrosis

Neonatal liver disease

Bile duct lesions.



Figure 20.8 Kayser–Fleischer rings from copper in the cornea in a child with Wilson disease.

reduce copper absorption. Pyridoxine is given to prevent peripheral neuropathy. Neurological improvement may take up to 12 months of therapy. About 30% of children with Wilson disease will die from hepatic complications if untreated. Liver transplantation is considered for children with acute liver failure or severe end-stage liver disease.

Fibropolycystic liver disease

This is a range of inherited conditions affecting the development of the intrahepatic biliary tree. Presentation is with liver and renal disease. The liver disease may include cystic disease of the liver or biliary tree or congenital hepatic fibrosis.

Congenital hepatic fibrosis (CHF) presents in children over 2 years old with hepatosplenomegaly, abdominal distension and portal hypertension. It differs from cirrhosis in that liver function tests are normal in the early stage. Liver histology shows large bands of hepatic fibrosis containing abnormal bile ductules. The consequent portal hypertension causes bleeding from varices. Cystic renal disease may coexist and may cause hypertension or renal dysfunction.

Non-alcoholic fatty liver disease

Non-alcoholic fatty liver disease (NAFLD) is the single most common cause of chronic liver disease in the developed world. It is a spectrum of disease, ranging from simple fatty deposition (steatosis) through to inflammation (steatohepatitis), fibrosis, cirrhosis and end-stage liver failure. The prognosis in childhood is uncertain; very few develop cirrhosis in childhood in contrast to 8–17% of adults. Most affected children are obese. They are usually asymptomatic, although some complain of vague right upper quadrant abdominal pain or lethargy. The diagnosis is often suspected following the incidental finding of an echogenic liver on ultrasound or mildly elevated transaminases carried out for some other reason. Liver biopsy demonstrates marked steatosis with or without inflammation or fibrosis. The pathogenesis is not fully understood but may be linked to insulin resistance. Treatment is targeted at weight loss, which may lead to liver function tests returning to normal.

Cirrhosis and portal hypertension

Cirrhosis is the end result of many forms of liver disease. It is defined pathologically as extensive fibrosis with regenerative nodules. It may be secondary to hepatocellular disease or to chronic bile duct obstruction (biliary cirrhosis). The main pathophysiological effects of cirrhosis are diminished hepatic function and portal hypertension with splenomegaly, varices and ascites (see Fig. 20.1). Hepatocellular carcinoma may develop.

Children with compensated cirrhosis may be asymptomatic if liver function is adequate. They will not be jaundiced and may have normal liver function tests. As the cirrhosis increases, however, the results of deteriorating liver function and portal hypertension become obvious (Fig. 20.9). Physical signs include palmar and plantar erythema and spider naevi, malnutrition and hypotonia. Dilated abdominal veins and splenomegaly suggest portal hypertension, although the liver may be impalpable.



Figure 20.9 Cirrhosis and portal hypertension. (i) Malnutrition with loss of fat and muscle bulk; (ii) distended abdomen from hepatosplenomegaly and ascites; (iii) scrotal swelling from ascites; and (iv) no jaundice, despite advanced liver disease.

Investigations include:

- Screening for the known causes of chronic liver disease (see Box 20.2)
- Upper gastrointestinal endoscopy to detect the presence of oesophageal varices and/or erosive gastritis
- Abdominal ultrasound – may show a shrunken liver and splenomegaly with gastric and oesophageal varices
- Liver biopsy – may be difficult because of increased fibrosis but may indicate the aetiology (e.g. typical changes in congenital hepatic fibrosis, copper storage).

As cirrhosis decompensates, biochemical tests may demonstrate an elevation of aminotransferases and alkaline phosphatase. The plasma albumin falls and the prothrombin time becomes increasingly prolonged.

Oesophageal varices

These are an inevitable consequence of portal hypertension and may develop rapidly in children. They are best diagnosed by upper gastrointestinal endoscopy, as a barium swallow may miss small varices. Acute bleeding is treated conservatively with blood transfusions and H_2-blockers (e.g. ranitidine) or omeprazole. If bleeding persists, octreotide infusion, vasopressin analogues, sclerotherapy or band ligation may be effective.

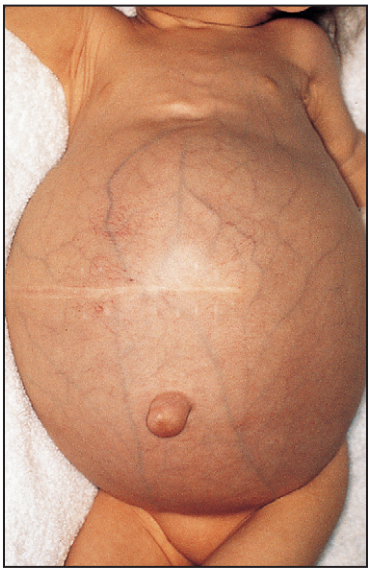


Figure 20.10 This infant has a grossly distended abdomen from ascites. There are dilated abdominal veins secondary to portal hypertension and an umbilical hernia from increased abdominal pressure. There is a surgical scar.

Portacaval shunts may preclude liver transplantation, but radiological placement of a stent between the hepatic and portal veins can be used as a temporary measure if transplantation is being considered.

Ascites

Ascites is a major problem (Fig. 20.10). The pathophysiology of ascites is uncertain, but contributory factors include hypoalbuminaemia, sodium retention, renal impairment and fluid redistribution. It is treated by sodium and fluid restriction and diuretics. Additional therapy for refractory ascites includes albumin infusions or paracentesis.

Spontaneous bacterial peritonitis

This should always be considered if there is undiagnosed fever, abdominal pain, tenderness or an unexplained deterioration in hepatic or renal function. A diagnostic paracentesis should be performed and the fluid sent for white cell count and differential and culture. Treatment is with broad-spectrum antibiotics.

Encephalopathy

This occurs in end-stage liver disease and may be precipitated by gastrointestinal haemorrhage, sepsis, sedatives, renal failure or electrolyte imbalance. It is difficult to diagnose in children as the level of consciousness may vary throughout the day. Infants present with irritability and sleepiness, while older children present with abnormalities in mood, sleep rhythm, intellectual performance and behaviour.



Figure 20.11 Many infants and children with liver disease need intensive nutritional supplementation. This malnourished infant is having both parenteral nutrition via a central line and continuous nasogastric feeding.

Plasma ammonia may be elevated and an EEG is always abnormal.

Renal failure

This may be secondary to renal tubular acidosis, acute tubular necrosis or functional renal failure (hepatorenal syndrome).

Management of children with liver disease

Although certain diseases may require specific treatment, the management of the sequelae of chronic liver disease is supportive, with the emphasis on correction of nutritional abnormalities, prevention of complications and intensive family support.

Nutrition

Malnutrition may be due to protein malnutrition, fat malabsorption, anorexia or fat-soluble vitamin deficiency (vitamins A, D, E and K).

Treatment is to provide a high-protein, high-carbohydrate diet with 50% more calories than the recommended dietary allowance. In children with cholestasis, medium-chain triglycerides, which are absorbed by the portal circulation, will provide fat, but 20–40% long-chain triglycerides are required to prevent essential fatty acid deficiency. Many children will require nasogastric tube feeding or parenteral nutrition (Fig. 20.11).

Fat-soluble vitamins

Vitamin K deficiency in liver disease may be due to malabsorption or diminished synthesis. Water-soluble forms of vitamin K are available.

Vitamin A deficiency causes night blindness in adults and retinal changes in infants. It is easily prevented with oral vitamin A.

Vitamin E deficiency causes peripheral neuropathy, haemolysis and ataxia. It is very poorly absorbed in cholestatic conditions and high oral doses are required.

Vitamin D deficiency causes rickets and pathological fractures. It is prevented by using a water-soluble form of vitamin D. Vitamin D-resistant rickets indicates renal tubular acidosis.

Pruritus

Many children with cholestasis have severe pruritus. Management is problematic but may include:

- Loose cotton clothing, avoiding overheating
- Emollients or evening primrose oil
- Phenobarbital to stimulate bile flow; cholestyramine which is a bile salt resin; ursodeoxycholic acid, an oral bile acid.

Encephalopathy

In children, encephalopathy is managed by treating the precipitating factor (sepsis, gastrointestinal haemorrhage) by protein restriction or by using oral lactulose to reduce ammonia reabsorption by lowering colonic pH and increasing colonic transit.

Liver transplantation

Liver transplantation is an accepted therapy for acute or chronic end-stage liver failure and has revolutionised the prognosis for these children. Transplantation is also considered for some hepatic malignancy.

The indications for transplantation in chronic liver failure are:

- Severe malnutrition unresponsive to intensive nutritional therapy
- Recurrent complications (bleeding varices, resistant ascites)
- Failure of growth and development
- Poor quality of life.

Liver transplant evaluation includes assessment of the vascular anatomy of the liver and exclusion of irreversible disease in other systems. Absolute contraindications include sepsis, untreatable cardiopulmonary disease or cerebrovascular disease.

There is considerable difficulty in obtaining small organs for children. Most children receive part of an adult's liver, either a cadaveric graft or more recently from a living related donor. A cadaveric organ may either be reduced to fit the child's abdomen (reduction hepatectomy) or split (shared between an adult and child).

Complications post-transplantation include:

- Primary non-function of the liver (5%)
- Hepatic artery thrombosis (10–20%)
- Biliary leaks and strictures (20%)
- Rejection (30–60%)
- Sepsis, the main cause of death.

In large national centres, the overall 1-year survival is approximately 90%, and the overall 5-year survival is >80%. Most deaths occur in the first 3 months. Children who survive the initial postoperative period usually do well. Long-term studies indicate normal psychosocial development and quality of life in survivors.

Further reading

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21

Malignant disease

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Cancer in children is not common.

- Around 1 child in 500 develops cancer by 15 years of age.
- Each year, in Western countries, there are 120–140 new cases per million children aged under 15 years, equivalent to about 1500 new cases each year in the UK.

The types of disease seen (Fig. 21.1) are very different from those in adults, where carcinomas of the lung, breast, gut and skin predominate. The age at presentation varies with the different types of disease:

- Leukaemia affects children at all ages (although there is an early childhood peak).
- Neuroblastoma and Wilms tumour are almost always seen in the first 6 years of life.
- Hodgkin lymphoma and bone tumours have their peak incidence in adolescence and early adult life.

Despite significant improvements in survival over the last four decades (Fig. 21.2), cancer is the commonest

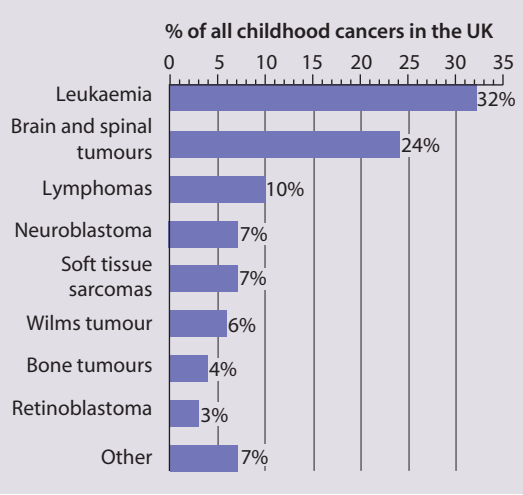


Figure 21.1 Relative frequency of different types of cancer in children in the UK.

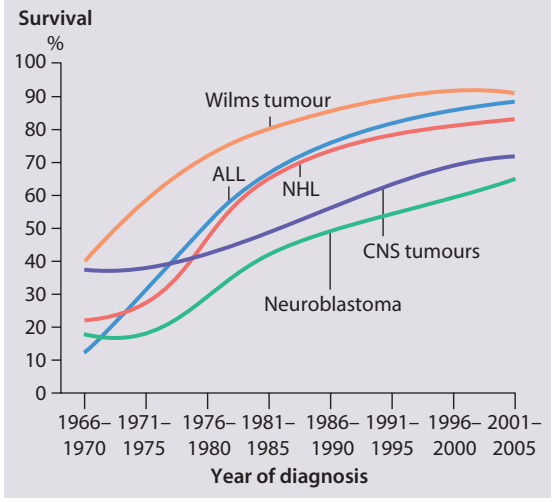


Figure 21.2 Five-year survival rates showing the considerable improvement over the last 50 years. (ALL, acute lymphoblastic leukaemia; NHL, non-Hodgkin lymphoma.) (Sources: 1971–2000. Stiller C (Editor) *Childhood Cancer in Britain – Incidence, Survival, Mortality*, Oxford, 2007, Oxford University Press; and 1966–1970 & 2001–2005, National Registry of Childhood Tumours, unpublished data.)

disease causing death in childhood (beyond the neonatal period). Overall, the 5-year survival of children with all forms of cancer is about 75%, most of whom can be considered cured, although cure rates vary considerably for different diagnoses. This improved life expectancy can be attributed mainly to the introduction of multi-agent chemotherapy, supportive care and specialist multidisciplinary management. However, for some children, the price of survival is long-term medical or psychosocial difficulties.

In children, leukaemia is the most common malignancy followed by brain tumours.

Aetiology

In most cases, the precise aetiology of childhood cancer is unclear, but it is likely to involve an interaction between environmental factors (e.g. viral infection) and host genetic susceptibility. In fact, there are very few established environmental risk factors, and although cancer occurs as a result of mutations in cellular growth controlling genes, which are usually sporadic but may be inherited, in most cases a specific gene mutation is unknown. One example of an inherited cancer is bilateral retinoblastoma, which is associated with a mutation within the RB gene located on chromosome 13. There is a wide range of syndromes associated with an increased risk of cancer in childhood, e.g. associations exist between Down syndrome and leukaemia, and neurofibromatosis and glioma. In time, the further identification of biological characteristics of specific tumour cells may also help elucidate the basic pathogenetic mechanisms behind their origin.

Clinical presentation

Cancer in children can present with:

- A localised mass
- The consequences of disseminated disease, e.g. bone marrow infiltration, causing systemic ill-health
- The consequences of pressure from a mass on local structures or tissue, e.g. airway obstruction secondary to enlarged lymph nodes in the mediastinum.

Investigations

Initial symptoms can be very non-specific and this can often lead to significant delays in diagnosis. Once a diagnosis of malignancy is suspected, the child should be referred to a specialist centre for further investigation.

Radiology

The location of solid tumours and evidence of any metastases are identified and localised, using a combination of ultrasound, plain X-rays, CT and MRI scans. Nuclear medicine imaging (e.g. radiolabelled technetium bone scan) may be useful to identify bone or bone marrow disease or, using special markers (MIBG scan), localise tumours of neural crest origin, e.g. neuroblastoma.

Tumour marker studies

Increased urinary catecholamine excretion (e.g. VMA, vanillylmandelic acid) is useful in confirming the diagnosis of neuroblastoma. High α-fetoprotein (αFP) production is often observed in germ cell tumours and liver tumours and can be used to monitor treatment response.

Pathology

All diagnoses must be confirmed histologically, either by bone marrow aspiration for cases of leukaemia or by biopsy for most solid tumours, although this may not always be possible for brain tumours. Histological techniques such as immunohistochemistry are routinely used to differentiate tumour types. Molecular and genetic techniques are also used to confirm diagnosis (e.g. translocation of chromosomes 11 and 22 in Ewing sarcoma) and to predict prognosis (e.g. amplification of the 'N-*myc*' oncogene associated with a poor prognosis in neuroblastoma).

Management

Once malignancy has been diagnosed, the parents and child need to be seen and the diagnosis explained to them in a realistic, yet positive way. Detailed investigation to define the extent of the disease (staging) is paramount to planning treatment. Children are usually treated as part of national and international collaborative studies that offer consistency in care and have contributed to improvements in outcome.

In the UK, children with cancer are initially investigated and treated in specialist centres where experienced multidisciplinary teams can provide the intensive medical and psychosocial support required. Subsequent management is often shared between the specialist centre, referral hospital and local services within the community, to provide the optimum care with the least disruption to the family.

Teenagers and young adults (TYA)

Survival statistics suggest that teenagers and young adults have poorer outcomes than children and constitute a distinct population. This relates both to the specific types and biological behaviour of their tumours and to their particular social/psychological needs. This has prompted the development of age-appropriate treatment protocols, facilities and support networks.

Treatment

Treatment may involve chemotherapy, surgery or radiotherapy, alone or in combination.

Chemotherapy

This is used:

- as primary curative treatment, e.g. in acute lymphoblastic leukaemia
- to control primary or metastatic disease before definitive local treatment with surgery and/or radiotherapy, e.g. in sarcoma or neuroblastoma
- as adjuvant treatment to deal with residual disease and to eliminate presumed micrometastases, e.g. after initial local treatment with surgery in Wilms tumour

The use of biologically targeted therapies and their role in combination with conventional treatment modalities is an area of active research that is likely to increase in importance in the near future.

Radiotherapy

This retains a role in the treatment of some tumours, but the risk of damage to growth and function of normal tissue is greater in a child than in an adult. The need for adequate protection of normal tissues and for careful positioning and immobilisation of the patient during treatment raises practical difficulties, particularly in young children.

Surgery

Initial surgery is frequently restricted to biopsy to establish the diagnosis, and more extensive operations are usually undertaken to remove residual tumour after chemotherapy and/or radiotherapy.

High-dose therapy with bone marrow rescue

The limitation of both chemotherapy and radiotherapy is the risk of irreversible damage to normal tissues, particularly bone marrow. Transplantation of bone marrow stem cells can be used as a strategy to intensify the treatment of patients with the administration of potentially lethal doses of chemotherapy and/or radiation. The source of the marrow stem cells may be allogeneic (from a compatible donor) or autologous (from the patient him/herself, harvested beforehand, while the marrow is uninvolved or in remission). Allogeneic transplantation is principally used in the management of high-risk or relapsed leukaemia and autologous stem cell support is used most commonly in the treatment of children with solid tumours whose prognosis is poor using conventional chemotherapy, e.g. advanced neuroblastoma.

Supportive care and side-effects of treatment

Cancer treatment produces frequent, predictable and often severe multisystem side-effects (Fig. 21.3). Supportive care is an important part of management and improvements in this aspect of cancer care have contributed to the increasing survival rates.

Short-term side-effects of chemotherapy

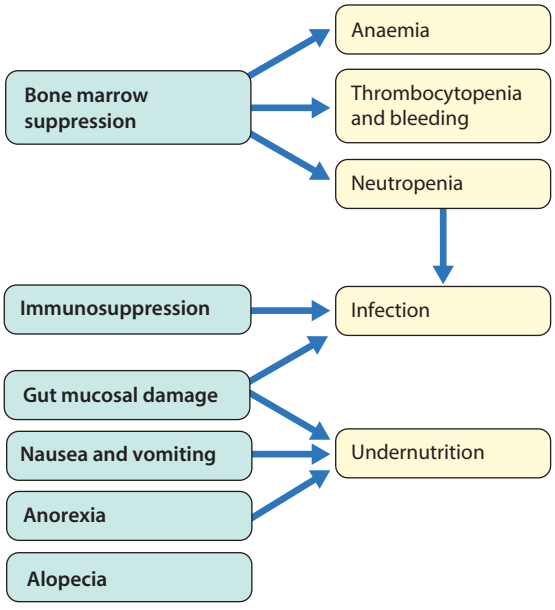


Figure 21.3 Short-term side-effects of chemotherapy.

Infection from immunosuppression

Due to both treatment (chemotherapy or wide-field radiation) and underlying disease, children with cancer are immunocompromised and at risk of serious infection. Children with fever and neutropenia must be admitted promptly to hospital for cultures and treatment with broad-spectrum antibiotics. Some important opportunistic infections associated with therapy for cancer include *Pneumocystis jiroveci* (*carinii*) pneumonia (especially in children with leukaemia), disseminated fungal infection (e.g. aspergillosis and candidiasis) and coagulase-negative staphylococcal infections of central venous catheters.

Most common viral infections are no worse in children with cancer than in other children, but measles and varicella zoster (chickenpox) may have atypical presentation and can be life-threatening. If non-immune, immunocompromised children are at risk from contact with measles or varicella, some protection can be afforded by prompt administration of immunoglobulin or zoster immune globulin. Aciclovir is used to treat established varicella infection, but no treatment is available for measles. During chemotherapy and from 6 months to a year subsequently, the use of live vaccines is contraindicated due to depressed immunity. After this period, re-immunisation against common childhood infections is recommended.

Bone marrow suppression

Anaemia may require blood transfusions. Thrombocytopaenia presents the hazard of bleeding, and considerable blood product support may be required, particularly for children with leukaemia, those undergoing intensive therapy requiring bone marrow

transplantation and in the more intensive solid tumour protocols.

Gastrointestinal damage, nausea and vomiting and nutritional compromise

Mouth ulcers are common, painful and, when severe, can prevent a child eating adequately. Many chemotherapy agents are nauseating and induce vomiting, which may be only partially prevented by the routine use of antiemetic drugs. These two complications can result in significant nutritional compromise. Chemotherapy-induced gut mucosal damage also causes diarrhoea and may predispose to Gram-negative infection.

Drug-specific side-effects

Many individual drugs have very specific side-effects: e.g. cardiotoxicity with doxorubicin; renal failure and deafness with cisplatin; haemorrhagic cystitis with cyclophosphamide; and neuropathy with vincristine. The extent of these side-effects is not always predictable and patients require careful monitoring during, and in some cases, after treatment is complete.

Other supportive care issues

Fertility preservation

Some patients may be at risk of infertility as a result of their cancer treatment. Appropriate fertility preservation techniques may involve surgically moving a testis or ovary out of the radiotherapy field; sperm banking (which should be offered to boys mature enough to achieve this); and consideration of newer techniques such as cryopreservation of ovarian cortical tissue, although the long-term efficacy of this is still uncertain.

Venous access

The discomfort of multiple venepunctures for blood sampling and intravenous infusions can be avoided with central venous catheters, although these do carry a risk of infection (Fig. 21.4).

Fever with neutropenia requires hospital admission, cultures and intravenous antibiotics.

Psychosocial support

The diagnosis of a potentially fatal illness has an enormous and long-lasting impact on the whole family. They need the opportunity to discuss the implications of the diagnosis and its treatment and their anxiety, fear, guilt and sadness. Most will benefit from the counselling and practical support provided by health professionals. Help with practical issues, including transport, finances, accommodation and care of siblings, is an early priority. The provision of detailed written material for parents will help them understand their child's disease and treatment. The children themselves, and their siblings, need an age-appropriate explanation of the disease. Once treatment is established and the disease appears to be under control, families should be encouraged to return to as normal a lifestyle as possible. Early return to school is important and children with cancer should not be allowed to under-achieve the expectations previously held for them. It is easy to underestimate the severe stress that persists within families in relation to the uncertainty of the long-term outcome. This often manifests itself as marital problems in parents and behavioural difficulties in both the child and siblings.



Figure 21.4 The central venous catheter allows pain-free blood tests and injections for this child on chemotherapy, which has caused the alopecia.

Summary

Malignant disease in children

- Uncommon, but affects 1 in 500 by 15 years of age
- The overall 5-year survival rate is 75%
- Presents with a localised mass or its pressure effects or disseminated disease
- Treatment may involve chemotherapy, surgery, radiotherapy or high-dose therapy with bone marrow rescue
- Fever with neutropenia must be investigated and treated urgently
- Measles and varicella zoster infection are potentially life-threatening
- Requires a multidisciplinary team to provide supportive care and psychosocial support
- Supportive care – includes not only management of side-effects but also pain management and fertility preservation
- Psychosocial support – includes not only the patient and parents but also siblings and other family and community members.

Leukaemia

Acute lymphoblastic leukaemia (ALL) accounts for 80% of leukaemia in children. Most of the remainder is acute myeloid/acute non-lymphocytic leukaemia (AML/ANLL). Chronic myeloid leukaemia and other myeloproliferative disorders are rare.

Clinical presentation

Presentation of acute lymphoblastic leukaemia peaks at 2–5 years. Clinical symptoms and signs result from disseminated disease and systemic ill-health from infiltration of the bone marrow or other organs with leukaemic blast cells (Fig. 21.5). In most children, leukaemia presents insidiously over several weeks (see Case History 21.1) but in some children the illness presents and progresses very rapidly.

Investigations

Full blood count – in most but not all children, the blood count is abnormal, with low haemoglobin, thrombocytopenia and evidence of circulating leukaemic blast cells. Bone marrow examination is essential to confirm the diagnosis and to identify immunological and cytogenetic characteristics which give useful prognostic information. Chest X ray is required to identify a mediastinal mass characteristic of T-cell disease.

Both ALL and AML are classified by morphology. Immunological phenotyping further subclassifies ALL; the common (75%) and T-cell (15%) subtypes are the most common. Prognosis and some aspects of clinical presentation vary according to different subtypes, and treatment intensity is adjusted accordingly.

Management of acute lymphoblastic leukaemia

A number of factors contribute to prognosis in ALL and dictate the intensity of therapy (Table 21.1).

A typical treatment schema is shown in Figure 21.7.

Remission induction

Before starting treatment of the disease, anaemia may require correction with blood transfusion, the risk of bleeding minimised by transfusion of platelets, and infection must be treated. Additional hydration and allopurinol (or urate oxidase when the white cell count is high and the risk is greater) are given to protect renal function against the effects of rapid cell lysis. Remission implies eradication of the leukaemic blasts and restoration of normal marrow function. Four weeks of

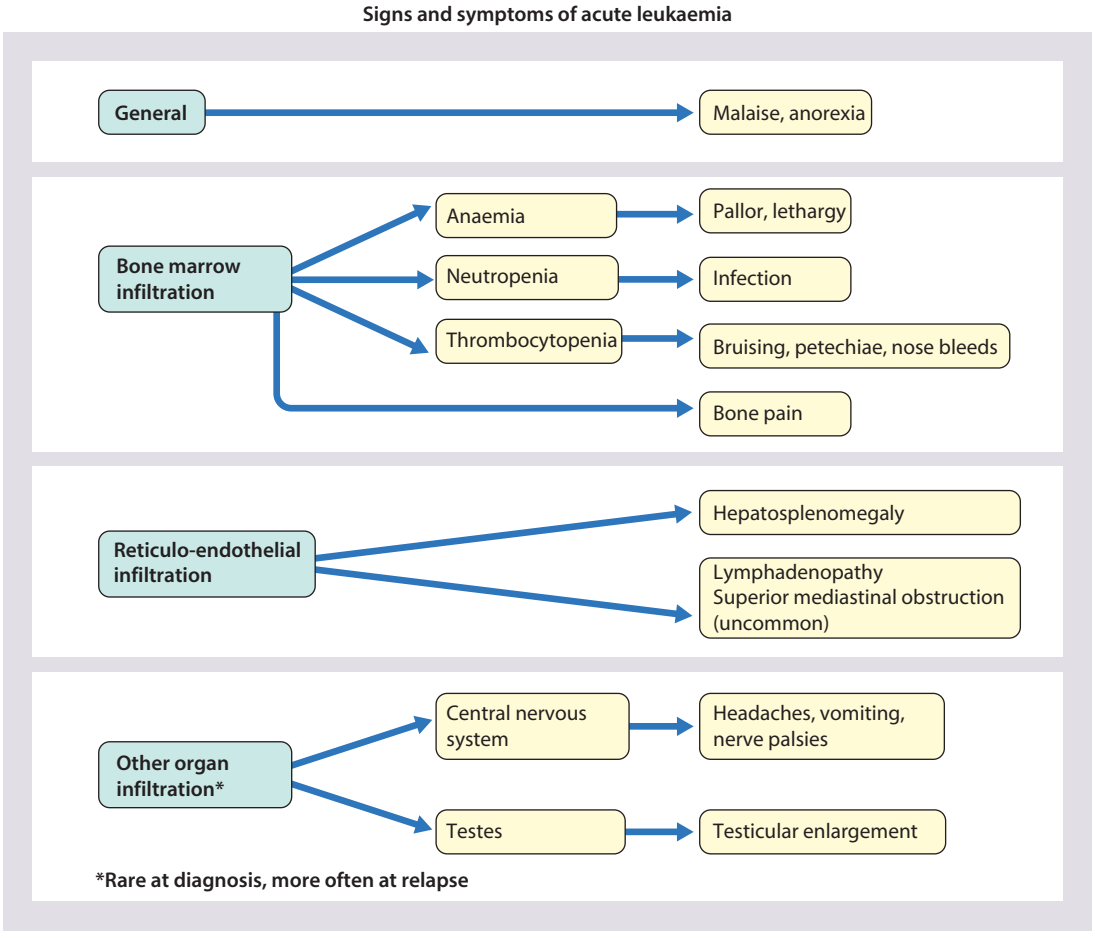


Figure 21.5 Signs and symptoms of acute leukaemia.

Case History

21.1 Disseminated disease, e.g. bone marrow infiltration, causing systemic ill-health

A 4-year-old girl was generally unwell, lethargic, looking pale and occasionally febrile over a period of 9 weeks. Two courses of antibiotics for recurrent sore throat failed to result in any benefit. Her parents returned to their general practitioner when she developed a rash. Examination showed pallor, petechiae, modest generalised lymphadenopathy and mild hepatosplenomegaly. A full blood count showed:

- Hb 8.3 g/dl
- WBC 15.6×10^9/L
- Platelets 44×10^9/L.

Blast cells were seen on the peripheral blood film. Cerebrospinal fluid (CSF) examination was normal. Bone marrow examination confirmed acute lymphoblastic leukaemia (Fig. 21.6).

Diagnosis: Acute lymphoblastic leukaemia.

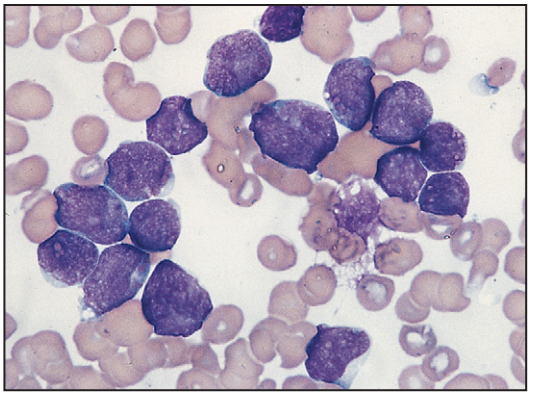


Figure 21.6 Leukaemic blast cells on a bone marrow smear.

Table 21.1 Prognostic factors in acute lymphatic leukaemia

Prognostic factor	High-risk features
Age	<1 year or >10 years
Tumour load (measured by the white cell count, WBC)	$>50 \times 10^9$/L
Cytogenetic/molecular genetic abnormalities in tumour cells	e.g. MLL rearrangement, t(4;11), hypodiploidy (<44 chromosomes)
Speed of response to initial chemotherapy	Persistence of leukaemic blasts in the bone marrow
Minimal residual disease assessment (MRD) (submicroscopic levels of leukaemia detected by PCR)	High

combination chemotherapy is given and current induction treatment schedules achieve remission rates of 95%.

Intensification

A block of intensive chemotherapy is given to consolidate remission. This improves cure rates but at the expense of increased toxicity.

Central nervous system

Cytotoxic drugs penetrate poorly into the CNS. As leukaemic cells in this site may survive effective systemic treatment, additional treatment with intrathecal chemotherapy is used to prevent CNS relapse.

Continuing therapy

Chemotherapy of modest intensity is continued over a relatively long period of time, up to 3 years from diagnosis. Co-trimoxazole prophylaxis is given routinely to prevent *Pneumocystis carinii* pneumonia.

Treatment of relapse

High-dose chemotherapy, usually with total body irradiation (TBI) and bone marrow transplantation, is used as an alternative to conventional chemotherapy after a relapse.

Brain tumours

In contrast to adults, brain tumours in children are almost always primary and 60% are infratentorial. They are the most common solid tumour in children and are the leading cause of childhood cancer deaths in the UK. The types of brain tumour are:

- Astrocytoma (~40%) – varies from benign to highly malignant (*glioblastoma multiforme*)
- Medulloblastoma (~20%) – arises in the midline of the posterior fossa. May seed through the CNS via the CSF and up to 20% have spinal metastases at diagnosis
- Ependymoma (~8%) – mostly in posterior fossa where it behaves like medulloblastoma
- Brainstem glioma (6%)
- Craniopharyngioma (4%) – a developmental tumour arising from the squamous remnant of

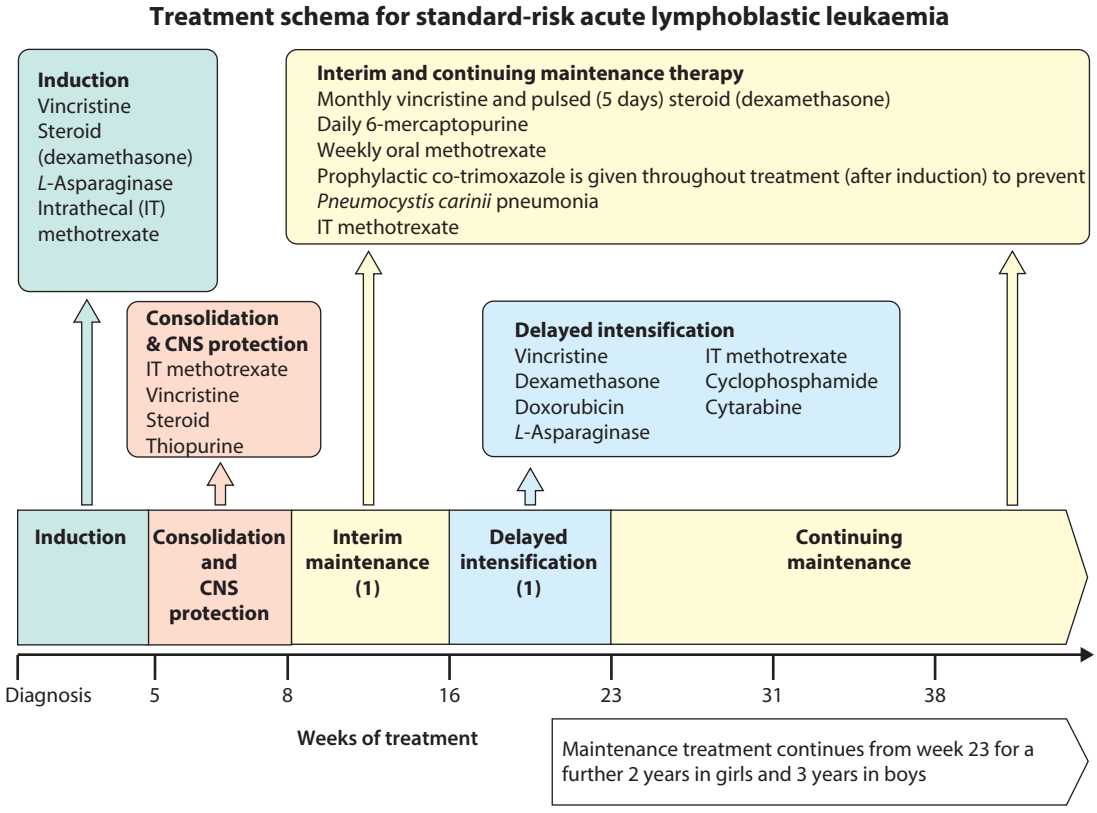


Figure 21.7 Treatment schema for standard-risk acute lymphoblastic leukaemia.

Rathke pouch. It is not truly malignant but is locally invasive and grows slowly in the suprasellar region.

Clinical features

Signs and symptoms are often related to evidence of raised intracranial pressure but focal neurological signs may be detected depending on the site of the tumour (see below). Spinal tumours, primary or metastatic, can present with back pain, peripheral weakness of arms or legs or bladder/bowel dysfunction, depending on the level of the lesion.

Persistent back pain in children warrants investigation with MRI scan.

Investigations

Brain tumours are best characterised on MRI scan. Magnetic resonance spectroscopy can be used to examine the biological activity of a tumour. Lumbar puncture must not be performed without neurosurgical advice if there is any suspicion of raised intracranial pressure.

Management

Surgery is usually the first treatment and is aimed at treating hydrocephalus, providing a tissue diagnosis and attempting maximum resection. In some cases the anatomical position of the tumour means biopsy is not safe, e.g. tumours in the brainstem. Even tumours which are histologically 'benign' can cause major challenges to survival. The use of radiotherapy and/or chemotherapy varies with tumour type and the age of the patient.

Late effects

The functional implications of the site of the tumour, the potential hazards of surgery and the importance of radiotherapy in treatment all combine to place children with brain tumours at particular risk of neurological disability and of growth, endocrine, neuropsychological and educational problems. Survivors may present complex combinations of these problems.

Lymphomas

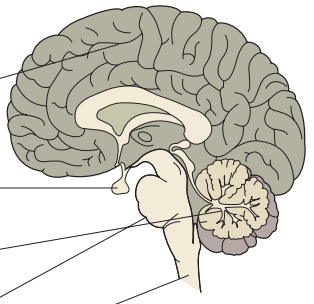
Lymphomas are malignancies of the cells of the immune system and can be divided into Hodgkin and non-Hodgkin lymphoma (NHL). NHL is more common in childhood, while Hodgkin lymphoma is seen more frequently in adolescence.

Hodgkin lymphoma

Clinical features

Classically presents with painless lymphadenopathy, most frequently in the neck. Lymph nodes are much larger and firmer than the benign lymphadenopathy

Brain tumours – clinical features



(a)

Raised intracranial pressure

Children and adolescents
Headache – worse in the morning
Vomiting – especially on waking in the morning
Behaviour/personality change
Visual disturbance
Papilloedema

Infants
Vomiting
Separation of sutures/tense fontanelle
Increased head circumference
Head tilt/posturing
Developmental delay/regression

Headaches and behaviour changes – is there raised intracranial pressure?

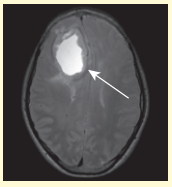
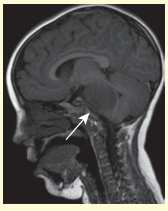
Site of tumour and clinical features specific to anatomical position	MRI Scans	Typical case history
Supratentorial – cortex Seizures Hemiplegia Focal neurological signs	 (b)	14 year old. Aggressive behaviour at school, headaches, seizure MRI scan – **(Fig 21.8b)** Diagnosis – **astrocytoma – glioblastoma multiforme** Management – surgery, radiotherapy +/- chemotherapy, but prognosis poor (<30% survival) Astrocytomas – commonly found in the cerebral hemispheres, thalamus and hypothalamus. For posterior fossa tumours, see below.
Midline Visual field loss – bitemporal hemianopia Pituitary failure – growth failure, diabetes insipidus, weight gain	 (c)	10 year old complaining of headaches, vomiting, poor growth, struggling to see the board at school. MRI scan – **(Fig 21.8c)** Diagnosis – **craniopharyngioma** Management – surgical excision +/- radiotherapy Prognosis – good survival but risk of long-term visual Impairment and lifelong, complex pituitary insufficiency
Cerebellar and IVth ventricle Truncal ataxia Coordination difficulties Abnormal eye movements	 (d)	3 year old vomiting in the mornings, unsteady on his feet, new-onset convergent squint. MRI scan – **(Fig 21.8d)** Diagnosis – **medulloblastoma** Management – surgery, chemotherapy, radiotherapy. Prognosis – survival rates are improving with 5-year survival about 50% Other posterior fossa tumours: Astrocytoma – cystic, slow growing. Good prognosis following surgery. Ependymoma – behaves like medulloblastoma
Brainstem Cranial nerve defects Pyramidal tract signs Cerebellar signs – ataxia Often no raised intracranial pressure	 (e)	4 year old. Refuses to walk, unable to climb stairs, squint, facial asymmetry and drooling. MRI scan – **(Fig 21.8e)** Diagnosis – **brainstem glioma.** But not for biopsy as too hazardous. Management – palliative radiotherapy Prognosis – very poor (<10% survival)

Figure 21.8 Location and clinical features of brain tumours. (a) Location of brain tumours. MRI scans showing (b) fronto-parietal mass, (c) large midline suprasellar mass, (d) cerebellar mass and (e) brainstem mass.

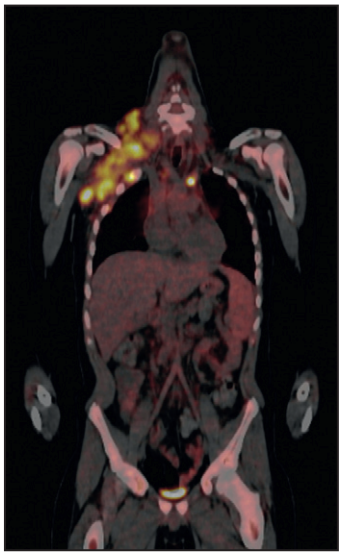


Figure 21.9 PET scan showing active disease in the right cervical and axillary nodes by uptake of FDG (an analogue of glucose).

commonly seen in young children. The lymph nodes may cause airways obstruction (see Case History 21.2). The clinical history is often long (several months) and systemic symptoms (sweating, pruritus, weight loss and fever – the so-called 'B' symptoms) are uncommon, even in more advanced disease.

Investigations

Lymph node biopsy, radiological assessment of all nodal sites and bone marrow biopsy is used to stage disease and determine treatment.

Management

Combination chemotherapy with or without radiotherapy. Positron emission tomography (PET) scanning is used in the UK to monitor treatment response and guide further management (Figure 21.9).

Overall, about 80% of all patients can be cured. Even with disseminated disease, about 60% can be cured.

Non-Hodgkin lymphoma

T-cell malignancies may present as acute lymphoblastic leukaemia or non-Hodgkin lymphoma, with both being characterised by a mediastinal mass with varying degrees of bone marrow infiltration. The mediastinal mass may cause superior vena caval obstruction. B-cell malignancies present more commonly as non-Hodgkin lymphoma, with localised lymph node disease usually in the head and neck or abdomen. Abdominal disease presents with pain from intestinal obstruction, a palpable mass or even intussusception in cases with involvement of the ileum.

Investigations

Biopsy, radiological assessment of all nodal sites (CT or MRI) and examination of the bone marrow and CSF.

Case History

21.2 Pressure from a mass on local structures or tissue, e.g. airway obstruction secondary to enlarged lymph nodes

A 14-year-old girl complained of a cough for 2 weeks which was non-productive and worse at night. She had seen her general practitioner and her chest was clear. She returned 2 weeks later, as she had noticed a swelling in her neck. On examination, she had a large anterior cervical lymph node which was non-tender. On referral to hospital, she had a chest X-ray, which showed a large mediastinal mass (Fig. 21.10).

Differential diagnosis

- T-cell non-Hodgkin lymphoma/acute lymphatic leukaemia
- Hodgkin lymphoma

Her full blood count was normal. A biopsy of the mass was consistent with a diagnosis of Hodgkin lymphoma.

Diagnosis: Hodgkin lymphoma.

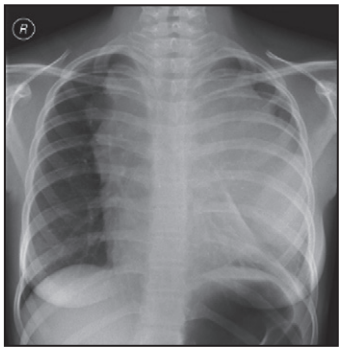


Figure 21.10 Chest X-ray showing a large mediastinal mass.

Management

Multi-agent chemotherapy. The majority of patients now do well and survival rates of over 80% are expected for both T- and B-cell disease.

Neuroblastoma

Neuroblastoma and related tumours arise from neural crest tissue in the adrenal medulla and sympathetic nervous system. It is a biologically unusual tumour in that spontaneous regression sometimes occurs in very young infants. There is a spectrum of disease from the benign (ganglioneuroma) to the highly malignant (neuroblastoma). Neuroblastoma is most common before the age of 5 years.

Clinical features

At presentation (Box 21.1), most children have an abdominal mass, but the primary tumour can lie anywhere along the sympathetic chain from the neck to the pelvis. Classically, the abdominal primary is of adrenal origin, but at presentation the tumour mass is often large and complex, crossing the midline and enveloping major blood vessels and lymph nodes. Paravertebral tumours may invade through the adjacent intervertebral foramen and cause spinal cord compression. Over the age of 2 years, clinical symptoms are mostly from metastatic disease, particularly bone pain, bone marrow suppression causing weight loss and malaise (see Case History 21.3).

Box 21.1 Presentation of neuroblastoma

Common	Less common
Pallor	Paraplegia
Weight loss	Cervical lymphadenopathy
Abdominal mass	Proptosis
Hepatomegaly	Periorbital bruising
Bone pain	Skin nodules
Limp	

Case History

21.3 Neuroblastoma

Jack, a 3-year-old boy, was taken to his general practitioner by his mother because he was not eating as well as usual and had a distended abdomen. Recently, he appeared reluctant to walk and sometimes cried when he was picked up. His grandmother thought he had lost weight. On examination, the general practitioner confirmed that he seemed generally miserable and pale. He was concerned to note a large abdominal mass. Urgent referral to his local hospital was made and, on arrival, he was also noted to be hypertensive.

Differential diagnosis and specific investigations

An initial ultrasound examination confirmed the abdominal mass and an MRI scan characterised a very large upper abdominal mass in complex relationship with the left kidney and the major vessels but extending towards the midline, suggestive of neuroblastoma (Fig. 21.11). Subsequent investigations confirmed bone marrow infiltration by tumour cells and a positive MIBG scan showing uptake at the primary and distant sites consistent with metastatic disease (Fig 21.12).

Diagnosis: Metastatic neuroblastoma.

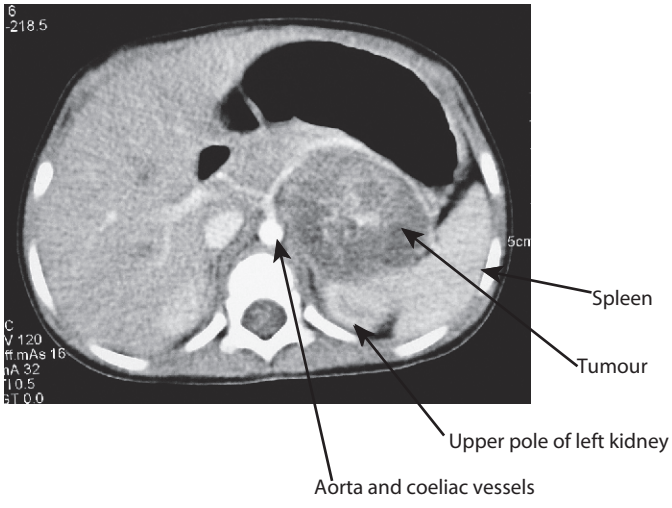


Figure 21.11 Transverse MRI image showing a large left-sided primary neuroblastoma arising from the adrenal region and distorting coeliac and mesenteric blood vessels.

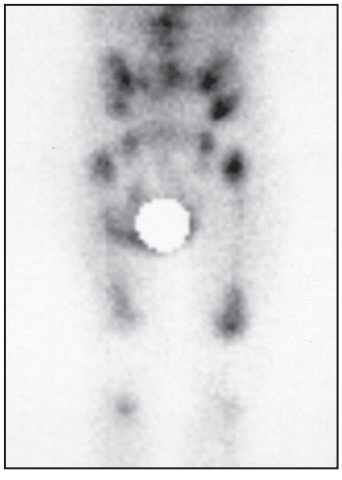


Figure 21.12 The MIBG scan 'maps' metastatic tumour marrow. This image shows the lower half of the abdomen, pelvis and legs. The dark areas are evidence of high isotope uptake and the pattern is consistent with widespread metastatic disease. (Normal uptake from excretion of isotope into urine in the bladder has been blocked in this exposure.)

Investigations

Characteristic clinical and radiological features with raised urinary catecholamine levels suggest neuroblastoma. Confirmatory biopsy is usually obtained and evidence of metastatic disease detected with bone marrow sampling, MIBG (metaiodobenzylguanidine) scan (as in Fig. 21.12) with or without a bone scan.

Age and stage of disease at diagnosis are the major factors which influence prognosis. Unfortunately, the majority of children over 1 year present with advanced disease and have a poor prognosis. Increasingly, information about the biological characteristics of neuroblastoma is being used to guide therapy and prognosis. Overexpression of the N-*myc* oncogene, evidence of deletion of material on chromosome 1 (del 1p) and gain of genetic material on chromosome 17q in tumour cells are all associated with a poorer prognosis.

Management

Localised primaries without metastatic disease can often be cured with surgery alone.

Metastatic disease is treated with chemotherapy, including high-dose therapy with autologous stem cell rescue, surgery and radiotherapy. Risk of relapse is high and the prospect of cure for children with metastatic disease is still little better than 30%. Immunotherapy and the use of long-term 'maintenance' treatment with differentiating agents (retinoic acid) are now establishing a role in those with high-risk disease.

Wilms tumour (nephroblastoma)

Wilms tumour originates from embryonal renal tissue and is the commonest renal tumour of childhood. Over 80% of patients present before 5 years of age and it is very rarely seen after 10 years of age.

Clinical features

Most children present with a large abdominal mass, often found incidentally in an otherwise well child. Other clinical features are listed in Box 21.2.

Investigations

Ultrasound and/or CT/MRI (Fig. 21.13) is usually characteristic, showing an intrinsic renal mass distorting the normal structure. Staging is to assess for distant metastases (usually in the lung), initial tumour resectability and function of the contralateral kidney.

Management

In the UK, children receive initial chemotherapy followed by delayed nephrectomy, after which the tumour is staged histologically and subsequent treatment is planned according to the surgical and pathological findings. Radiotherapy is restricted to those with more advanced disease.

Prognosis is good, with more than 80% of all patients cured. Cure rate for patients with metastatic disease at presentation (~15%) is over 60%, but relapse carries a poor prognosis.

Soft tissue sarcomas

Rhabdomyosarcoma is the most common form of soft tissue sarcoma in childhood. The tumour is thought to originate from primitive mesenchymal tissue and there are a wide variety of primary sites, resulting in varying presentations and prognosis.

Box 21.2 Presentation of Wilms tumour

Common	Uncommon
Abdominal mass	Abdominal pain
	Anorexia
	Anaemia (haemorrhage into mass)
	Haematuria
	Hypertension

About 5% have bilateral disease at diagnosis.

Wilms tumour

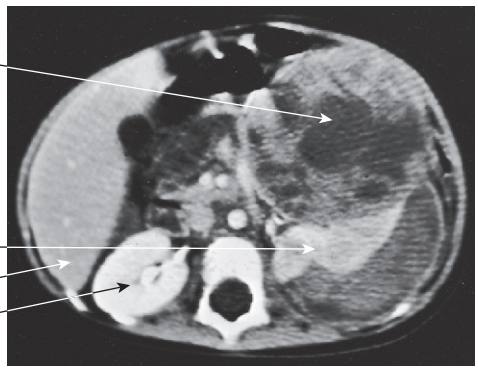
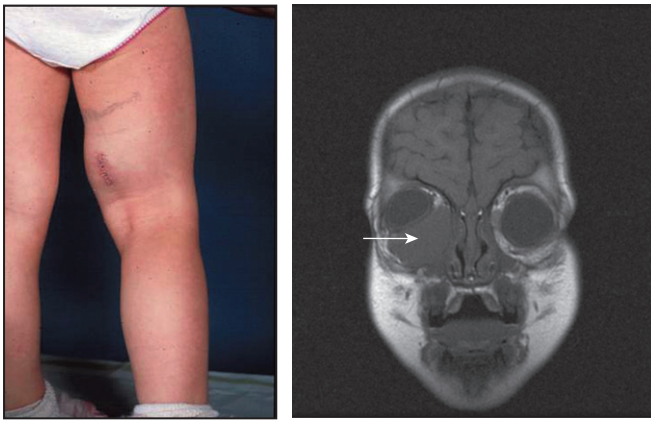


Figure 21.13 Large Wilms tumour arising within the left kidney, showing characteristic cystic and solid tissue densities.



(a) (b)

Figure 21.14 Rhabdomyosarcoma. **(a)** Soft tissue mass of lower limb. The scar is from a biopsy. **(b)** MRI scan of a child presenting with proptosis of the right eye. It shows a right periorbital soft tissue mass displacing the globe and compressing other orbital structures. Histology confirmed the diagnosis of rhabdomyosarcoma.

Clinical features

Head and neck are the most common sites of disease (40%), causing, e.g. proptosis, nasal obstruction or bloodstained nasal discharge.

Genitourinary tumours may involve the bladder, paratesticular structures or the female genitourinary tract. Symptoms include dysuria and urinary obstruction, scrotal mass or bloodstained vaginal discharge.

Metastatic disease (lung, liver, bone or bone marrow) is present in approximately 15% of patients at diagnosis and is associated with a particularly poor prognosis.

Investigations

Biopsy and full radiological assessment of primary disease and any evidence of metastasis (Fig. 21.14).

Management

Multimodality treatment (chemotherapy, surgery and radiotherapy) is used, dependent on the age of the patient and the site, size and extent of disease. The tumour margins are deceptively ill-defined, and attempts at primary surgical excision are often unsuccessful and are not attempted unless this can be achieved without mutilation or irreversible organ damage. Overall cure rates are about 65%.

Bone tumours

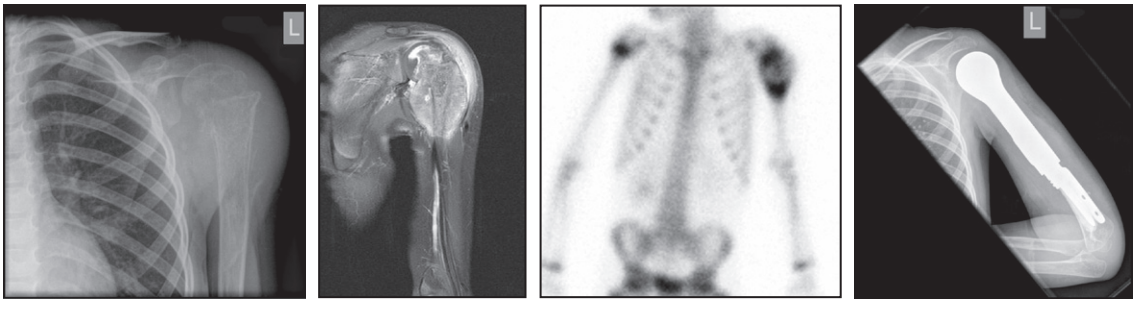
Malignant bone tumours are uncommon before puberty. Osteogenic sarcoma is more common than Ewing sarcoma, but Ewing sarcoma is seen more often in younger children. Both have a male predominance.

Clinical features

The limbs are the most common site. Persistent localised bone pain is the characteristic symptom, usually preceding the detection of a mass, and is an indication for early X-ray. At diagnosis, most patients are otherwise well.

Investigations

Plain X-ray is followed by MRI and bone scan (Fig. 21.15a–c). A bone X-ray shows destruction and variable periosteal new bone formation. In Ewing sarcoma, there is often a substantial soft tissue mass. Chest CT is used to assess for lung metastases and bone marrow sampling to exclude marrow involvement.



(a) (b) (c) (d)

Figure 21.15 Ewing sarcoma of the humerus. **(a)** Plain X-ray shows a destructive bone lesion within the proximal humeral metaphysis. **(b)** MRI shows a large destructive soft tissue mass arising from the proximal metadiaphysis of the left humerus. **(c)** Bone scan shows prominent abnormal tracer uptake in the proximal left humerus. **(d)** Post-surgery, most of the humerus has been resected and replaced by a metallic prosthesis.

Management

In both tumours, treatment involves the use of combination chemotherapy given before surgery. Whenever possible, amputation is avoided by using *en bloc* resection of tumours with endoprosthetic resection (Fig. 21.15d). In Ewing sarcoma, radiotherapy is also used in the management of local disease, especially when surgical resection is impossible or incomplete, e.g. in the pelvis or axial skeleton.

Retinoblastoma

Retinoblastoma is a malignant tumour of retinal cells and, although rare, it accounts for about 5% of severe visual impairment in children. It may affect one or both eyes. All bilateral tumours are hereditary, as are about 20% of unilateral cases. The retinoblastoma susceptibility gene is on chromosome 13, and the pattern of inheritance is dominant, but with incomplete penetrance. Most cases present within the first 3 years of life. Children from families with the hereditary form of the disease should be screened regularly from birth.

Clinical features

The most common presentation of unsuspected disease is when a white pupillary reflex is noted to replace the normal red one (Fig. 21.16) or with a squint.

Investigations

MRI and examination under anaesthetic. Tumours are frequently multifocal.

Treatment

The aim is to cure, yet preserve vision. Biopsy is not undertaken and treatment is based on the ophthalmological findings. Enucleation of the eye may be necessary for more advanced disease. Chemotherapy is used, particularly in bilateral disease, to shrink the tumour(s), followed by local laser treatment to the retina. Radiotherapy may be used in advanced disease, but it is more often reserved for the treatment of recurrence. Most patients are cured, although many are visually impaired. There is a significant risk of second malignancy (especially sarcoma) among survivors of hereditary retinoblastoma.



Figure 21.16 White pupillary reflex in retinoblastoma.

Rare tumours

Liver tumours

Primary malignant liver tumours are mostly hepatoblastoma (65%) or hepatocellular carcinoma (25%). Presentation is usually with abdominal distension or with a mass. Pain and jaundice are rare. Elevated serum α-fetoprotein (αFP) is detected in nearly all cases of hepatoblastoma and in some cases of hepatocellular carcinoma. Management includes chemotherapy, surgery and, in inoperable cases, liver transplantation is required. The majority of children with hepatoblastoma can now be cured, but the prognosis for children with hepatocellular carcinoma is less satisfactory.

Germ cell tumours

Germ cell tumours (GCTs) may be benign or malignant. They arise from the primitive germ cells which migrate from yolk sac endoderm to form gonads in the embryo. Benign tumours are most common in the sacrococcygeal region, and most malignant germ cell tumours are found in the gonads. Serum markers (αFP and β-HCG) are invaluable in confirming the diagnosis and in monitoring response to treatment. Malignant germ cell tumours are very sensitive to chemotherapy, and a very good outcome can be expected for disease at most sites, including the brain.

Langerhans cell histiocytosis

Langerhans cell histiocytosis (LCH) is a rare disorder characterised by an abnormal proliferation of histiocytes. It is no longer believed to be a truly malignant condition and is classified as a disorder of dendritic (antigen presenting) cells. However, its sometimes aggressive behaviour and its response to chemotherapy place it within the practice of oncologists. Clinically, its manifestations include:

- Bone lesions – present at any age with pain, swelling or even fracture. X-ray reveals a characteristic lytic lesion with a well-defined border, often involving the skull (Fig. 21.17).



Figure 21.17 Lytic bone lesions on a skull X-ray in Langerhans cell histiocytosis.

- Diabetes insipidus – may be associated with skull disease with proptosis and hypothalamic infiltration
- Systemic LCH – the most aggressive form which tends to present in infancy with a seborrhoeic rash (Fig. 21.18) and soft tissue involvement of the gums, ears, lungs, liver, spleen, lymph nodes and bone marrow. This form of LCH is usually progressive and requires chemotherapy, although spontaneous regression may occur.

The prognosis is variable, but most patients are cured.

Long-term survivors

Improved survival rates means an ever-increasing population of adult survivors of childhood cancer. Over half have at least one residual problem as a consequence of either the disease or its treatment (Table 21.2).

All survivors need regular long-term follow-up to provide appropriate treatment or advice. This need for specialist multidisciplinary follow-up continues into adulthood, and its provision presents a challenge within adult healthcare services. Until recently, the majority of survivors have remained under the care of paediatric oncologists, although specialist adult clinics are being established. Some survivors will require specific counselling for problems such as poor or asymmetric growth, infertility and sexual dysfunction, and advances in the use of adult growth hormone. Assisted conception techniques have enhanced the lives of many patients. The risk of second cancer is small, but nevertheless survivors are at increased risk and this may rise with increasing survival rates. When new treatment protocols for childhood cancers are developed, there is a need to reduce, whenever possible, the toxicity of treatment to spare the children adverse short- and long-term effects.



Figure 21.18 Rash in systemic Langerhans cell histiocytosis. It is often mistaken for seborrhoeic dermatitis or eczema.

Palliative care

When a child relapses, further treatment may be considered. A reasonable number can still be cured and others may have a further significant remission with good-quality life. However, for some children, a time comes when death is inevitable and the staff and family must make the decision to concentrate on palliative care.

Most parents prefer to care for their terminally ill child at home, but will need practical help and emotional support. Pain control and symptom relief are a serious source of anxiety for parents, but they can often be achieved successfully at home. Health professionals with experience in palliative care for children work together with the family and local healthcare workers. After the child's death, families should be offered continuing contact with an appropriate member of the team who looked after their child, and be given support through their bereavement.

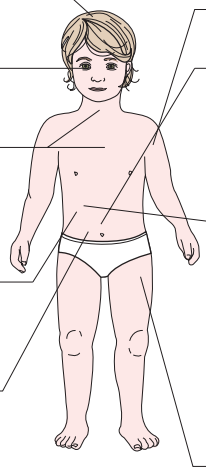
With adequate support from health professionals, palliative care for children can usually be provided at home.

Table 21.2 Some problems that may occur following cure of childhood cancer

Problem	Cause
Specific organ dysfunction	Nephrectomy for Wilms tumour
	Toxicity from chemotherapy, e.g. renal from cisplatin or ifosfamide, cardiac from doxorubicin or mediastinal radiotherapy
Growth/endocrine problems	Growth hormone deficiency from pituitary irradiation
	Bone growth retardation at sites of irradiation
Infertility	Gonadal irradiation
	Alkylating agent chemotherapy (cyclophosphamide, ifosfamide)
Neuropsychological problems	Cranial irradiation (particularly at age <5 years)
	Brain surgery
Second malignancy	Irradiation
	Alkylating agent chemotherapy
Social/educational disadvantage	Chronic ill health
	Absence from school

Summary

Presentation of malignant disease in children



Brain tumours:
- Raised intracranial pressure
- Neurological signs – depends on anatomical position

Retinoblastoma:
- Screening if positive family history
- White pupillary reflex or squint

Lymphomas:
- Enlarged lymph nodes in the head and neck or abdomen
- Mediastinal mass – may cause superior vena caval obstruction.

Wilms tumour:
- Large abdominal mass in a well child
- Occasionally anorexia, abdominal pain, haematuria

Langerhans cell histiocytosis:
- Seborrhoeic rash
- Widespread soft tissue infiltration
- Bone pain, swelling or fracture
- Diabetes insipidus

Soft tissue sarcomas:
- Mass any site

Neuroblastoma:
- Abdominal mass, crosses the midline
- Spinal cord compression
- Weight loss and malaise
- Pallor, bruising
- Bone pain

Acute lymphoblastic leukaemia (ALL):
- Malaise, anorexia
- Pallor, lethargy
- Infections
- Bruising, petichiae, nose bleeds
- Lymphadenopathy
- Hepatosplenomegaly
- Bone pain

Malignant bone tumours:
- Localised bone pain

Pre-school (<5 years old)	School-aged	Adolescence
Acute lymphoblastic leukaemia (ALL) – peak incidence Non-Hodgkin lymphoma	Acute lymphoblastic leukaemia (ALL)	Acute lymphoblastic leukaemia (ALL) Hodgkin lymphoma
Neuroblastoma	Brain tumours	Malignant bone tumours
Wilm tumour		Soft tissue sarcomas
Retinoblastoma		

Further reading

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Short textbook

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Comprehensive textbook

Pizzo PA, Poplack DG, editors: *Principles and Practice of Pediatric Oncology*, ed 6, Lippincott, 2010, Williams and Wilkins.

Comprehensive textbook

Stevens MCG, Caron HN, Biondi A, editors: *Cancer in Children: Clinical Management*, ed 6, Oxford, 2011, Oxford University Press.

Short textbook

Stiller C, editor: *2007 Childhood Cancer in Britain: Incidence, Survival, Mortality*, Oxford, 2007, Oxford University Press.

Websites (Accessed May 2011)

CCLG (Children Cancer and Leukaemia Group): Available at: http://www.cclg.org.uk/index.php
Association of healthcare professionals involved in the treatment and care of children and younger teenagers with cancer, underpins all the activity in paediatric oncology in the UK.

Cure4Kids. Available at: http://www.cure4kids.org
Dedicated to improving healthcare for children with cancer and other catastrophic diseases around the globe. Provides continuing medical education and communication tools to healthcare professionals and scientists worldwide.

22

Haematological disorders

Anaemia 382
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Thrombosis in children 402

Haemopoiesis is the process which maintains lifelong production of haemopoietic (blood) cells. The main site of haemopoiesis in fetal life is the liver, whereas throughout postnatal life, it is the bone marrow. All haemopoietic cells are derived from *pluripotent haemopoietic stem cells*, which are crucial for normal blood production; deficiency causes bone marrow failure because stem cells are required for the ongoing replacement of dying cells. Haemopoietic stem cells can also be used for treatment, e.g. cells from healthy donors can be transplanted into children with bone marrow failure (*stem cell transplantation*).

Haemoglobin production in the fetus and newborn

The most important difference between haemopoiesis in the fetus compared with postnatal life is the changing pattern of haemoglobin (Hb) production at each stage of development. The composition and names of these haemoglobins are shown in Table 22.1. Understanding the developmental changes in haemoglobin helps to explain the patterns of abnormal haemoglobin production in some inherited childhood anaemias. Embryonic haemoglobins (Hb Gower 1, Hb Gower 2 and Hb Portland) are produced between 4 and 8 weeks' gestation, after which haemoglobin production switches to fetal haemoglobin (HbF). HbF is made up of 2 α chains and 2 γ chains ($\alpha_2\gamma_2$) and is the main Hb during fetal life. HbF has a higher affinity for oxygen than adult Hb (HbA), and is therefore better able to hold on to oxygen, an advantage in the relatively hypoxic environment of the fetus (Fig. 22.1). At birth, the types of Hb are: HbF, HbA and HbA_2. HbF is gradually replaced by HbA and HbA_2 during the first year of life. By 1 year of age, the percentage of HbF is very low in healthy children and increased proportions of HbF are a sensitive indicator of some inherited disorders of haemoglobin production (haemoglobinopathies) .

Haematological values at birth and the first few weeks of life

Features are:

- At birth, the Hb in term infants is high, 14–21.5 g/dl, to compensate for the low oxygen concentration in the fetus. The Hb falls over the first few weeks, mainly due to reduced red cell production, reaching a nadir of around 10 g/dl at 2 months of age (Fig. 22.2). Normal haematological

Table 22.1 Embryonic, fetal and adult haemoglobins

Haemoglobin type	Globin chains	
	α-gene cluster	β-gene cluster
Embryonic		
Hb Gower 1	ξ_2	ε_2
Hb Gower 2	α_2	ε_2
Hb Portland	ξ_2	γ_2
Fetal		
HbF	α_2	γ_2
Adult		
HbA	α_2	β_2
HbA_2	α_2	δ_2
Haemoglobin types in newborns and adults		
Newborn	HbF 74%, HbA 25%, HbA_2 1%	
Children >1 year old and adults	HbA 97%, HbA_2 2%	

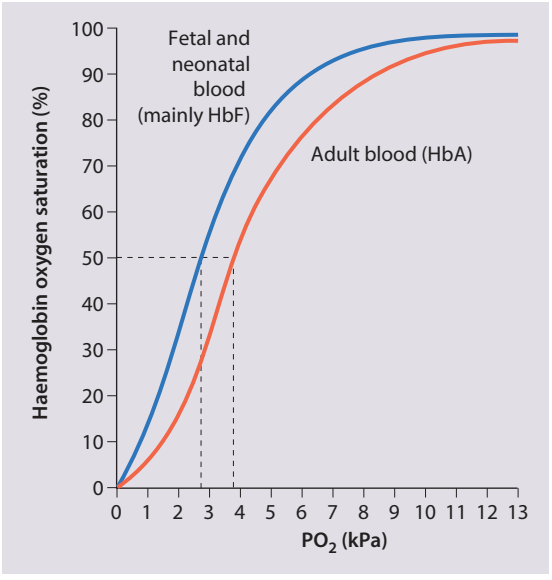


Figure 22.1 Oxygen dissociation curve showing the left shift of HbF compared with HbA. HbF-containing red cells have a higher affinity for oxygen and hold on to oxygen, delivering less to the tissues.

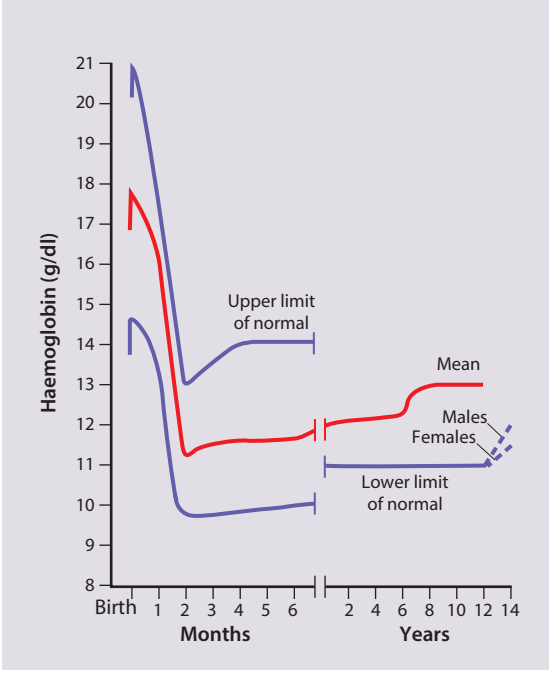


Figure 22.2 Changes in haemoglobin concentration with age, showing that the haemoglobin is high at birth, falling to its lowest concentration at 2–3 months of age.

values at birth and during childhood are shown in the Appendix.

- Preterm babies have a steeper fall in Hb to a mean of 6.5–9 g/dl at 4–8 weeks chronological age.
- Normal blood volume at birth varies with gestational age. In healthy term infants the average blood volume is 80 ml/kg; in preterm infants the average blood volume is 100 ml/kg.
- Stores of iron, folic acid and vitamin B_{12} in term and preterm babies are adequate at birth. However, in preterm infants, stores of iron and folic acid are lower and are depleted more quickly, leading to deficiency after 2–4 months if the recommended daily intakes are not maintained by supplements.
- White blood cell counts in neonates are higher than in older children ($10–25 \times 10^9$/L).
- Platelet counts at birth are within the normal adult range ($150–400 \times 10^9$/L).

Summary

Haemoglobin at birth

- The Hb concentration is high at birth (>14 g/dl) but falls to its lowest level at 2 months of age.
- Fetal Hb (HbF) is gradually replaced by adult Hb (HbA + HbA_2) during infancy.

Anaemia

Anaemia is defined as an Hb level below the normal range. The normal range varies with age, so anaemia can be defined as:

- Neonate: Hb <14 g/dl
- 1–12 months: Hb <10 g/dl
- 1–12 years: Hb <11 g/dl.

Anaemia results from one or more of the following mechanisms:

- Reduced red cell production – either due to ineffective erythropoiesis (e.g. iron deficiency, the commonest cause of anaemia) or due to red cell *aplasia*
- Increased red cell destruction (*haemolysis*)
- Blood loss – relatively uncommon cause in children.

There may be a combination of these three mechanisms, e.g. *anaemia of prematurity*.

Using this approach, the principal causes of anaemia are shown in Figure 22.3 and a diagnostic approach to identifying their causes is shown in Figure 22.4.

The definition of anaemia varies with age: Hb <10 g/dl in infants (post-neonatal), Hb <11 g/dl from 1 to 12 years old.

Anaemia due to reduced red cell production

Reduced red cell production may be due to:

- 'Ineffective erythropoiesis': here red cell production occurs at a normal or increased rate but differentiation or survival of the red cells is defective (e.g. iron deficiency).
- Complete absence of red cell production (red cell aplasia).

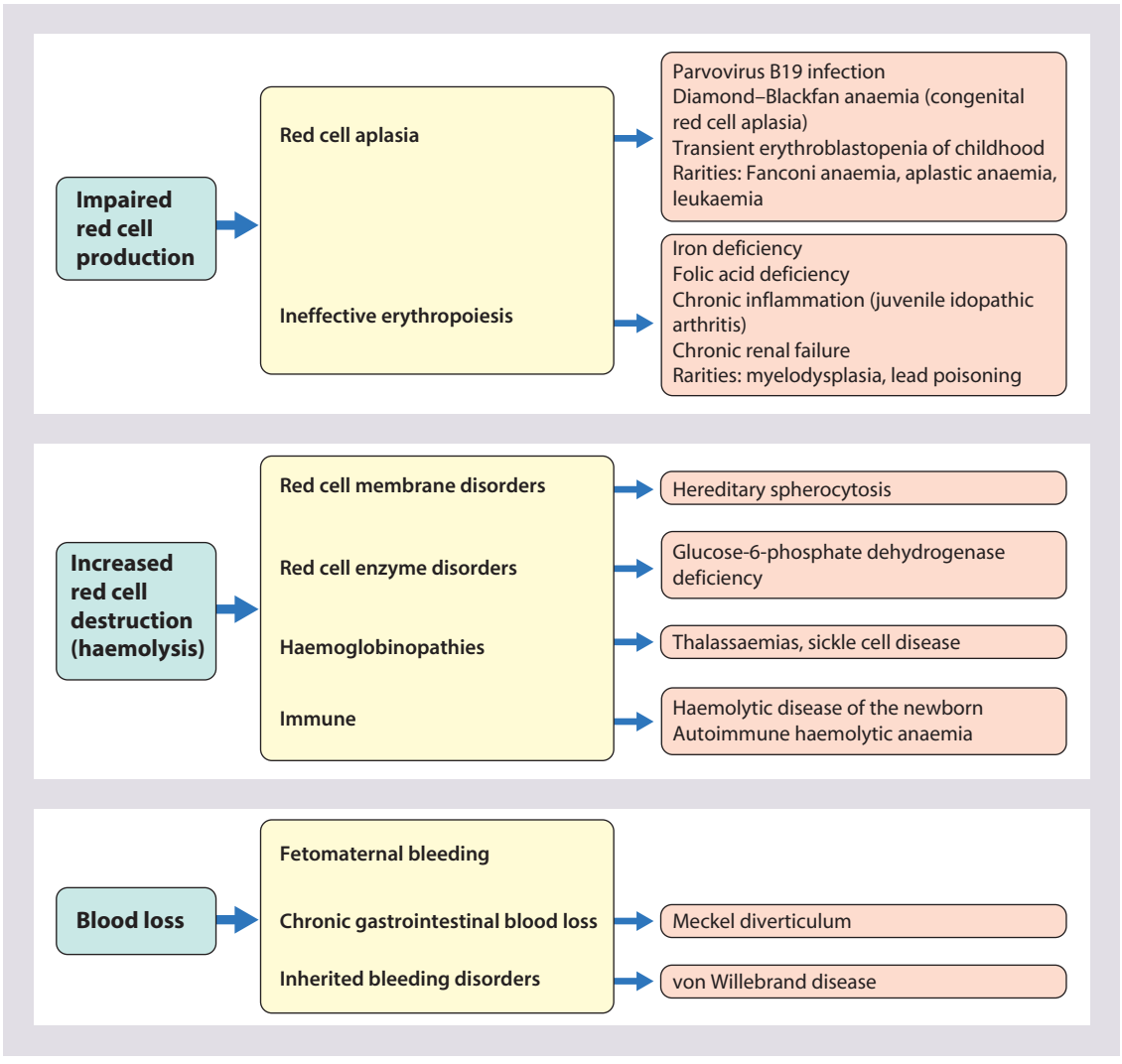


Figure 22.3 Causes of anaemia in infants and children.

Causes of anaemia in infants and children

Diagnostic clues to ineffective erythropoiesis are:

- Normal reticulocyte count
- Abnormal mean cell volume (MCV) of the red cells: low in iron deficiency and raised in folic acid deficiency.

Iron deficiency

The main causes of iron deficiency are:

- Inadequate intake
- Malabsorption
- Blood loss.

Inadequate intake of iron is common in infants because additional iron is required for the increase in blood volume accompanying growth and to build up the child's iron stores (Fig. 22.5). A 1-year-old infant requires an intake of iron of about 8 mg/day, which is about the same as his father (9 mg/day) but only half that of his mother (15 mg/day).

Iron may come from:

- Breast milk (low iron content but 50% of the iron is absorbed)
- Infant formula (supplemented with adequate amounts of iron)
- Cow's milk (higher iron content than breast milk but only 10% is absorbed)
- Solids introduced at weaning, e.g. cereals (cereals are supplemented with iron but only 1% is absorbed).

Iron deficiency may develop because of a delay in the introduction of mixed feeding beyond 6 months of age or to a diet with insufficient iron-rich foods, especially if it contains a large amount of cow's milk (Box 22.1). Iron absorption is markedly increased when eaten with food rich in vitamin C (fresh fruit and vegetables) and is inhibited by tannin in tea.

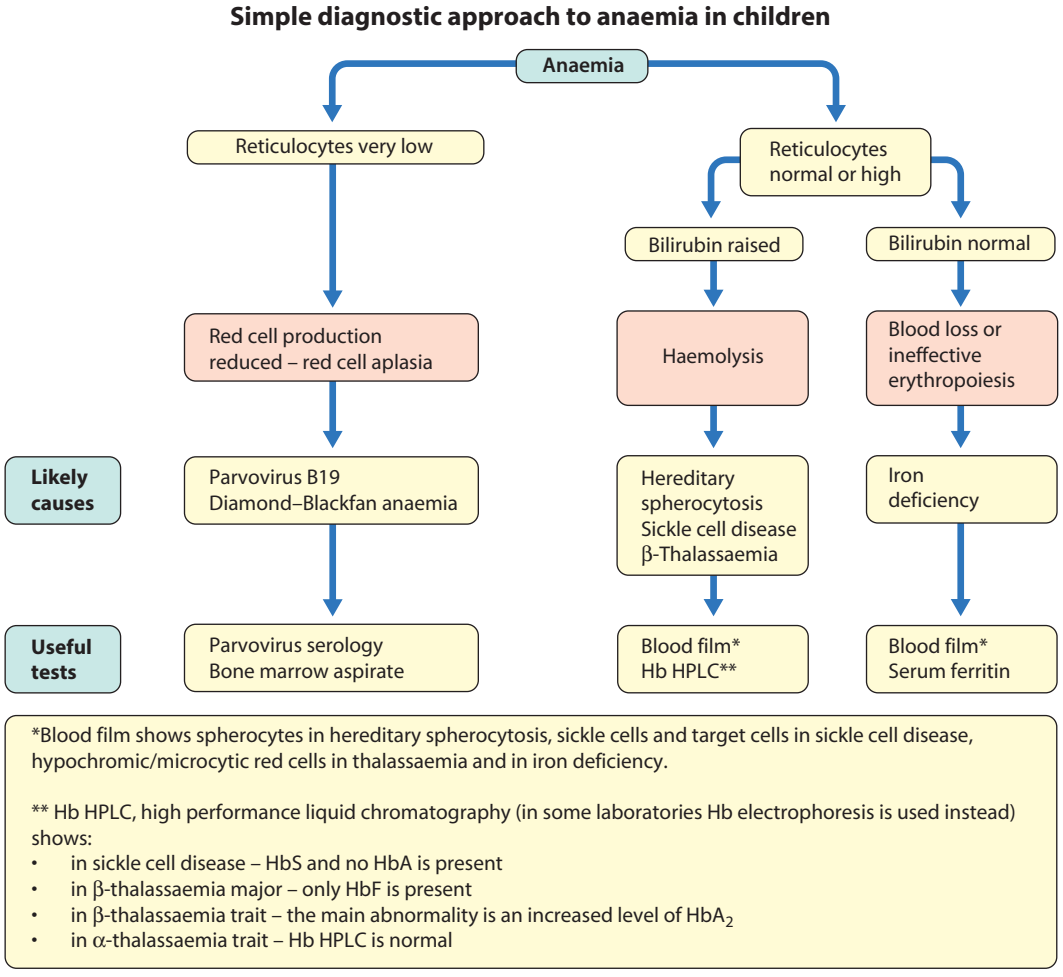


Figure 22.4 Diagnostic approach to anaemia.

Box 22.1 Dietary sources of iron

High in iron
- Red meat – beef, lamb
- Liver, kidney
- Oily fish – pilchards, sardines, etc.

Average iron
- Pulses, beans and peas
- Fortified breakfast cereals with added vitamin C
- Wholemeal products
- Dark green vegetables – broccoli, spinach, etc.
- Dried fruit – raisins, sultanas
- Nuts and seeds – cashews, peanut butter, etc.

Foods to avoid in excess in toddlers
- Cow's milk
- Tea: tannin inhibits iron uptake
- High-fibre foods: phytates inhibit iron absorption.

Infants should not be fed unmodified cow's milk as its iron content is low and poorly absorbed.

Clinical features

Most infants and children are asymptomatic until the Hb drops below 6–7 g/dl. As the anaemia worsens, children tire easily and young infants feed more slowly than usual. The history should include asking about blood loss and symptoms or signs suggesting malabsorption. They may appear pale but pallor is an unreliable sign unless confirmed by pallor of the conjunctivae, tongue or palmar creases. Some children have 'pica', a term which describes the inappropriate eating of non-food materials such as soil, chalk, gravel or foam rubber (see Case History 22.1). There is evidence that iron deficiency anaemia may be detrimental to behaviour and intellectual function.

Iron requirements during childhood

Diagnosis

The diagnostic clues are:

- Microcytic, hypochromic anaemia (low MCV and MCH)
- Low serum ferritin.

The other main causes of microcytic anaemia are:

- β-thalassaemia trait (usually children of Asian, Arabic or Mediterranean origin)

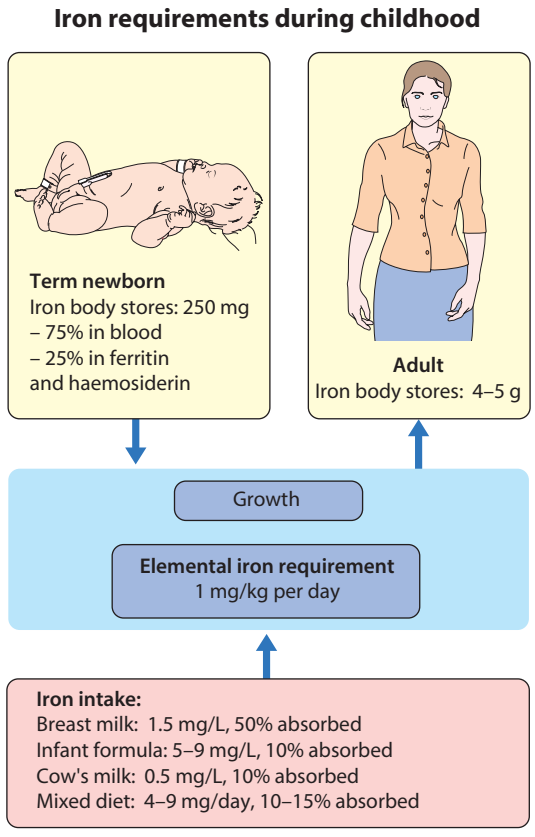


Figure 22.5 Iron requirements during childhood.

Case History

22.1 Iron deficiency anaemia

Ayesha, aged 2 years, was noted to look pale when she attended her general practitioner for an upper respiratory tract infection. A blood count showed Hb 5.0 g/dl, MCV 54 fl (normal 72–85 fl) and MCH 16 (normal 24–39 pg). She was drinking 3 pints of cow's milk per day and was a very fussy eater, refusing meat. She had started eating soil when playing in the garden.

Because of the inappropriately large volume of milk she was drinking, she was not sufficiently hungry to eat solid food. Replacing some of the milk with iron-rich food and treatment with oral iron produced a rise in the Hb to 7.5 g/dl within 4 weeks. Her pica (eating non-food materials) stopped. Oral iron was continued until her Hb had been normal for 3 months.

- α-thalassaemia trait (usually children of African or Far Eastern origin)
- Anaemia of chronic disease (e.g. due to renal failure).

Management

For most children, management involves *dietary advice* and supplementation with *oral iron*. The best tolerated preparations are Sytron (sodium iron edetate) or Niferex (polysaccharide iron complex) – unlike some other preparations these do not stain the teeth. Iron supplementation should be continued until the Hb is normal and then for a minimum of a further 3 months to replenish the iron stores. With good compliance, the Hb will rise by about 1 g/dl per week. Failure to respond to oral iron usually means the child is not getting the treatment. However, investigation for other causes, in particular malabsorption (e.g. due to coeliac disease) or chronic blood loss (e.g. due to Meckel diverticulum) is advisable if the history or examination suggests a non-dietary cause or if there is failure to respond to therapy in compliant patients. Blood transfusion should never be necessary for dietary iron deficiency. Even children with an Hb as low as 2–3 g/dl due to iron deficiency have arrived at this low level over a prolonged period and can tolerate it.

Treatment of iron deficiency with normal Hb

Some children have biochemical evidence of iron deficiency (e.g. low serum ferritin) but have not yet developed anaemia. Whether these children should be treated with oral iron is controversial. In favour of treatment is the knowledge that iron is required for normal brain development and there is evidence that iron deficiency anaemia is associated with behavioural and intellectual deficiencies, which may be reversible with iron therapy. However, it is not yet clear whether treatment of subclinical iron deficiency confers significant benefit. Treatment also carries a risk of accidental poisoning with oral iron, which is very toxic. A simple strategy is to provide dietary advice to increase oral iron and its absorption in all children with subclinical deficiency and to offer parents the option of additional treatment with oral iron supplements.

Treatment of iron deficiency anaemia is with dietary advice and oral iron therapy for several months.

Red cell aplasia

There are three main causes of red cell aplasia in children:

- Congenital red cell aplasia ('Diamond–Blackfan anaemia')
- Transient erythroblastopenia of childhood
- Parvovirus B19 infection (this infection only causes red cell aplasia in children with inherited haemolytic anaemias and not in healthy children).

The diagnostic clues to red cell aplasia are:

- Low reticulocyte count despite low Hb
- Normal bilirubin
- Negative direct antiglobulin test (Coombs test)
- Absent red cell precursors on bone marrow examination.

Diamond–Blackfan anaemia (DBA) is a rare disease (5–7 cases/million live births). There is a family history in 20% of cases; the remaining 80% are sporadic. Specific